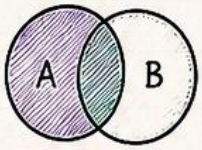
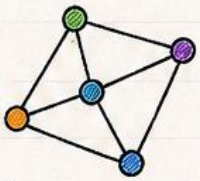
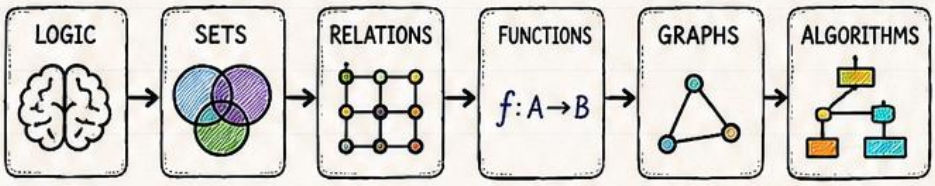
DISCRETE MATHEMATICS NOTES





FROM BASICS TO ADVANCED



- Understand.
- Practice.
- Solve.
- Master.
- Score High.

TOP 100

★ IMPORTANT QUESTIONS ★

+

QUICK REVISION CHEAT SHEET

TOPICS COVERED

1. Logic & Propositional Logic
2. Sets & Set Operations
3. Relations & Functions
4. Mathematical Induction
5. Recurrence Relations
6. Counting Principles
7. Permutations & Combinations
8. Pigeonhole Principle
9. Graph Theory
10. Trees & Spanning Trees
11. Boolean Algebra
12. Algebraic Structures
13. Number Theory & Modular Arithmetic
14. More Important Problems
15. Top 100 Exam Questions

+ Cheat Sheet for Quick Revision!



> "Mathematics is the language of logic, patterns, and possibilities.
> Master Discrete Math,
> Master Problem Solving."

★

WHAT YOU'LL GAIN

- ☑ Strong Conceptual Clarity
- ☑ Faster Problem Solving
- ☑ Exam & Interview Confidence
- ☑ Quick Revision Anytime
- ☑ Better Logic & Analytical Thinking
- ☑ Higher Accuracy in Coding & CS





1. Mathematical Logic & Propositions

What is Mathematical Logic ?

Mathematical logic is the study of statements (propositions) and their logical relationships using symbols and rules of inference.

What is a Proposition ?

A proposition is a declarative sentence that is either True (T) or False (F), but not both.

Examples :

- 2 + 2 = 4 (True)
- C# is a programming language. (True)
- Paris is the capital of France. (True)
- 5 is an even number. (False)
- What time is it ? (Not a proposition)

> **Note :**
> A proposition must be a statement. It cannot be a question, command or an exclamation.

Logical Connectives

Symbol	Name	Meaning
$\sim p$	NOT	Negation
$p \wedge q$	AND	Conjunction
$p \vee q$	OR	Disjunction
$p \rightarrow q$	IMPLIES	Implication
$p \leftrightarrow q$	IFF	Biconditional

Truth Tables

NOT

p	$\sim p$
T	F
F	T

AND ($\wedge$)

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

OR ($\vee$)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Implication ($\rightarrow$)

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional ($\leftrightarrow$)

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Laws of Logic

- Identity Law : $p \wedge T \equiv p$, $p \vee F \equiv p$
- Domination Law : $p \wedge F \equiv F$, $p \vee T \equiv T$
- Idempotent Law : $p \wedge p \equiv p$, $p \vee p \equiv p$
- Double Negation Law : $\sim\sim p \equiv p$
- Commutative Law : $p \wedge q \equiv q \wedge p$, $p \vee q \equiv q \vee p$
- Associative Law : $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
  $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- Distributive Law : $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
  $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
- De Morgan's Laws : $\sim (p \wedge q) \equiv \sim p \vee \sim q$
  $\sim (p \vee q) \equiv \sim p \wedge \sim q$

Compound Propositions

If p and q are propositions, then we can form new propositions using logical connectives.

Example :

- p : It is raining.
- q : I have an umbrella.

Then,

- $p \wedge q$: It is raining and I have an umbrella.
- $p \vee q$: It is raining or I have an umbrella.
- $p \rightarrow q$: If it is raining, then I have an umbrella.
- $p \leftrightarrow q$: It is raining if and only if I have an umbrella.

Tautology, Contradiction & Contingency

Tautology	:	A proposition that is always True. Example : $p \vee \sim p$
Contradiction	:	A proposition that is always False. Example : $p \wedge \sim p$
Contingency	:	A proposition that is sometimes True and sometimes False. Example : $p \rightarrow q$

2. Predicate Logic & Quantifiers

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What is Predicate Logic ?

Predicate Logic (also called First-Order Logic) extends propositional logic by allowing statements about objects using predicates, variables, quantifiers and functions. It helps us express general statements and relations.

Note :

- Predicate Logic deals with statements that contain variables.
- Propositional Logic deals with whole statements.

Basic Terms

Predicate	A statement about objects. Example : P(x) → "x is Prime"
Variable	A symbol that represents an object. Example : x, y, z
Constant	A fixed value. Example : 2, 5, Delhi
Domain	The set of all possible objects. Example : Integers ($\mathbb{Z}$), Natural numbers (N)
Universe of Discourse	The complete set under consideration. Example : All students in a class

Syntax of Predicate Logic

P(x)	:	Predicate P holds for x
¬P(x)	:	Not P(x)
P(x) ∧ Q(x)	:	P(x) and Q(x)
P(x) ∨ Q(x)	:	P(x) or Q(x)
P(x) → Q(x)	:	If P(x) then Q(x)
P(x) ↔ Q(x)	:	P(x) if and only if Q(x)
∀x P(x)	:	For all x, P(x) is true
∃x P(x)	:	There exists an x such that P(x) is true

Examples of Predicate Logic Statements

① P(x) : x is a human.
② Q(x) : x is a student.
③ R(x, y) : x is the parent of y.
④ ∀x (P(x) → Q(x))
"Every human is a student."
⑤ ∃x (P(x) ∧ ¬Q(x))
"There exists a human who is not a student."
⑥ ∀x ∃y R(x, y)
"Every person has at least one parent."

Order of Quantifiers Matters

① ∀x ∃y P(x, y) ≠ ∃y ∀x P(x, y)
"For every x, there exists a y such that P(x, y) is true." "There exists a y, such that for every x, P(x, y) is true."

② ∀x ∀y P(x, y) ≠ ∀y ∀x P(x, y)
(In general, we cannot change the order of quantifiers with different variables.)

Quantifiers

Symbol	Name	Meaning	Example
∀	Universal Quantifier	For all / for every	∀x P(x) "For all x, P(x) is true"
∃	Existential Quantifier	There exists / there is at least one	∃x P(x) "There exists an x such that P(x) is true"

Examples

- ∀x (x > 0) → "For all x, x is greater than 0."
- ∃x (x^2 = 4) → "There exists an x such that x^2 = 4."
- ∀x ∃y (x < y) → "For every x, there exists a y such that x < y."
- ∃x ∀y (x + y = y + x) → "There exists an x such that for all y, x + y = y + x."

Negation with Quantifiers

Statement	Negation
∀x P(x)	∃x ¬P(x) "Not for all x, P(x)" = "There exists an x such that P(x) is not true"
∃x P(x)	∀x ¬P(x) "Not there exists an x, P(x)" = "For all x, P(x) is not true"

Common Rules / Equivalences

① ¬∀x P(x) ≡ ∃x ¬P(x)
② ¬∃x P(x) ≡ ∀x ¬P(x)
③ ∀x (P(x) ∧ Q(x)) ≡ (∀x P(x)) ∧ (∀x Q(x))
④ ∃x (P(x) ∨ Q(x)) ≡ (∃x P(x)) ∨ (∃x Q(x))
⑤ ∀x (P(x) → Q(x)) ≡ ∃x (P(x) ∧ ¬Q(x)) with ¬ outside
¬∀x (P(x) → Q(x)) ≡ ∃x (P(x) ∧ ¬Q(x))
⑥ ∃x (P(x) → Q(x)) ≡ ∀x (P(x) → Q(x))
(when domain is non-empty)

3. Sets & Set Operations

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What is a Set ?

A set is a well-defined collection of distinct objects, called elements or members.
Sets are usually denoted by uppercase letters:
A, B, C, ...

Example :
A = {1, 2, 3, 4}
B = {3, 4, 5}

Note :
- Order does not matter.
- No duplicate elements.

Basic Notations

$x \in A$	x is an element of set A
$x \notin A$	x is not an element of set A
$A \subseteq B$	A is a subset of B
$A \subset B$	A is a proper subset of B ($A \subseteq B$ and $A \neq B$)
$A \supseteq B$	A is a superset of B
$A = B$	A is equal to B
$\|A\|$	Cardinality (number of elements) in set A

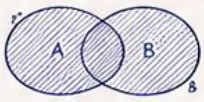
Types of Sets

Type	Description & Example
Empty Set (Null Set)	A set with no elements. $\emptyset$ or $\{\}$
Singleton Set	A set with exactly one element. $\{a\}$
Finite Set	A set with finite number of elements. $\{1, 2, 3, 4\}$
Infinite Set	A set with infinite number of elements. $\{1, 2, 3, \ldots\}$
Equal Sets	Two sets are equal if they have exactly the same elements. $\{1,2\} = \{2,1\}$
Equivalent Sets	Two sets are equivalent if they have the same number of elements. $\{a,b,c\}$ and $\{1,2,3\}$

Set Operations

1. Union ($A \cup B$)

Elements in A or B or both.

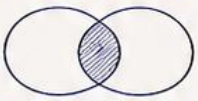


$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Example :
$A = \{1,2,3\}$, $B = \{3,4,5\}$
$A \cup B = \{1,2,3,4,5\}$

2. Intersection ($A \cap B$)

Elements common in A and B.

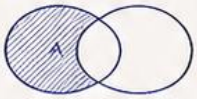


$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Example :
$A = \{1,2,3\}$, $B = \{3,4,5\}$
$A \cap B = \{3\}$

3. Difference ($A - B$)

Elements in A but not in B.

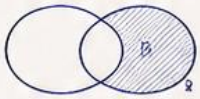


$A - B = \{x \mid x \in A \text{ and } x \notin B\}$

Example :
$A = \{1,2,3\}$, $B = \{3,4,5\}$
$A - B = \{1,2\}$

4. Difference ($B - A$)

Elements in B but not in A.

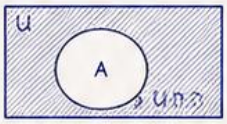


$B - A = \{x \mid x \in B \text{ and } x \notin A\}$

Example :
$A = \{1,2,3\}$, $B = \{3,4,5\}$
$B - A = \{4,5\}$

5. Complement (A^c)

Elements not in A (w.r.t Universal Set U).

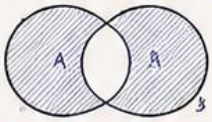


$A^c = \{x \mid x \in U \text{ and } x \notin A\}$

Example :
$U = \{1,2,3,4,5\}$, $A = \{1,2\}$
$A^c = \{3,4,5\}$

6. Symmetric Difference ($A \Delta B$)

Elements in A or B but not in both.



$A \Delta B = (A - B) \cup (B - A)$

Example :
$A = \{1,2,3\}$, $B = \{3,4,5\}$
$A \Delta B = \{1,2,4,5\}$

7. Universal Set (U)

The set of all objects under consideration.



Example :
$U = \{1,2,3,4,5,6\}$

8. Cartesian Product ($A \times B$)

Set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

$A \times B = \{(a,b) \mid a \in A, b \in B\}$

Example :
$A = \{1,2\}$, $B = \{x,y\}$
$A \times B = \{(1,x), (1,y), (2,x), (2,y)\}$

Important Laws

① $A \cup \emptyset = A$
② $A \cap \emptyset = \emptyset$
③ $A \cup U = U$
④ $A \cap U = A$
⑤ $A \cup A = A$
⑥ $A \cap A = A$
⑦ $A \cup A^c = U$
⑧ $A \cap A^c = \emptyset$
⑨ $(A \cup B) = (B \cup A)$ Commutative
⑩ $(A \cap B) = (B \cap A)$ Commutative
⑪ $(A \cup B) \cup C = A \cup (B \cup C)$ Associative
⑫ $(A \cap B) \cap C = A \cap (B \cap C)$ Associative
⑬ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ Distributive
⑭ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ Distributive
⑮ $A - B = A \cap B^c$
⑯ $(A - B) \cup B = A \cup B$
⑰ $(A \cap B)^c = A^c \cup B^c$ (De Morgan's Law)
⑱ $(A \cup B)^c = A^c \cap B^c$ (De Morgan's Law)

Quick Facts

- ☑ Number of subsets of a set with n elements = 2^n
- ☑ Number of proper subsets = $2^n - 1$
- ☑ Number of non-empty subsets = $2^n - 1$
- ☑ Number of ways to choose r elements from n = $n_{Cr} = \frac{n!}{r!\,(n-r)!}$
- ☑ If $|A| = m$ and $|B| = n$, then $|A \times B| = m \times n$

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4. Relations & Properties

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• What is a Relation ?

A relation from a set A to a set B is a subset of $A \times B$.

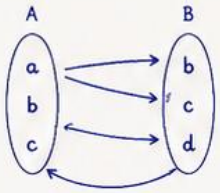
If A = B, then the relation is said to be on set A.

Note :
- Every relation is a set of ordered pairs.
- Relations can have any number of elements.

• Types of Relations

Type	Description
Empty Relation	$R = \emptyset$ (no ordered pair)
Universal Relation	$R = A \times B$ (all possible pairs)
Identity Relation	$I_A = \{(a,a) \mid a \in A\}$
Inverse Relation	$R^{-1} = \{(b,a) \mid (a,b) \in R\}$
Equality Relation	$R = I_A$ (identity relation)
Complement of R	$\overline{R} = (A \times B) - R$

• Representation of Relations

1. Set of Ordered Pairs	2. Arrow Diagram	3. Matrix Form
$R = \{(a,b), (a,c), (b,c), (c,a)\}$ from A to B where $A = \{a, b, c\}$ $B = \{b, c, d\}$		$A \setminus B$ | b c d a | 1 1 0 b | 0 1 0 c | 1 0 0 (1 → related, 0 → not related)

• Properties of Relations (on a set A)

Property	Definition	Symbolic Form	Example
Reflexive	Every element is related to itself.	$\forall\, a \in A,\ (a,a) \in R$	$R = \{(1,1), (2,2), (3,3), (1,2)\}$ is reflexive.
Irreflexive	No element is related to itself.	$\forall\, a \in A,\ (a,a) \notin R$	$R = \{(1,2), (2,3)\}$ is irreflexive.
Symmetric	If a is related to b, then b is related to a.	$\forall\, a,b \in A,$ $(a,b) \in R \rightarrow (b,a) \in R$	$R = \{(1,2), (2,1), (2,3), (3,2)\}$ is symmetric.
Antisymmetric	If a is related to b and b is related to a, then a = b.	$\forall\, a,b \in A,$ $(a,b) \in R$ and $(b,a) \in R$ $\rightarrow a = b$	$R = \{(1,1), (1,2), (2,2), (2,3), (3,3)\}$ is antisymmetric.
Asymmetric	If a is related to b, then b is not related to a.	$\forall\, a,b \in A,$ $(a,b) \in R \rightarrow (b,a) \notin R$	$R = \{(1,2), (2,3), (3,4)\}$ is asymmetric.
Transitive	If a is related to b and b is related to c, then a is related to c.	$\forall\, a,b,c \in A,$ $(a,b) \in R$ and $(b,c) \in R$ $\rightarrow (a,c) \in R$	$R = \{(1,2), (2,3), (1,3)\}$ is transitive.

• Equivalence Relation

A relation R on a set A is an equivalence relation if it is

1. Reflexive
2. Symmetric
3. Transitive

Equivalence relations divide a set into equivalence classes.

Example :

Let $A = \{1, 2, 3, 4, 5, 6\}$

Define $R = \{(a,b) \mid a \equiv b \pmod 2\}$

Then R is an equivalence relation.

Equivalence classes :

$[1] = \{1, 3, 5\}$, $[2] = \{2, 4, 6\}$

• Partial Order Relation (Poset)

A relation R on a set A is a partial order if it is

1. Reflexive
2. Antisymmetric
3. Transitive

Example :

Let $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (1,3), (2,3), (2,4), (3,4)\}$

("$\leq$" relation on natural numbers)

Then R is a partial order.

• Composition of Relations

Let R be a relation from A to B and S be a relation from B to C. The composition of R and S (denoted by $S \circ R$) is a relation from A to C defined as :

$S \circ R = \{(a,c) \mid \exists\, b \in B$ such that $(a,b) \in R$ and $(b,c) \in S\}$

• Properties at a Glance

- ☑ Reflexive + Symmetric + Transitive ⇒ Equivalence Relation
- ☑ Reflexive + Antisymmetric + Transitive ⇒ Partial Order
- ☑ Symmetric + Antisymmetric ⇒ Only Identity Relation
- ☑ Asymmetric ⇒ Irreflexive
- ☑ Antisymmetric ⇒ Not necessarily Asymmetric

① Check if R is Reflexive on $A = \{1,2,3\}$
$R = \{(1,1), (2,2), (3,3), (1,2)\}$
Since $(1,1), (2,2), (3,3) \in R$
→ R is Reflexive.

② Check if R is Symmetric
$R = \{(1,2), (2,1), (2,3), (3,2)\}$
For every $(a,b), (b,a) \in R$
→ R is Symmetric.

③ Check if R is Transitive
$R = \{(1,2), (2,3), (1,3)\}$
$(1,2)$ and $(2,3) \rightarrow (1,3) \in R$
→ R is Transitive.

④ Check if R is Antisymmetric
$R = \{(1,1), (1,2), (2,2), (2,3), (3,3)\}$
No distinct a,b have (a,b) and (b,a)
→ R is Antisymmetric.

5. Functions & Types

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What is a Function?

A function from a set A to a set B is a rule that assigns to each element of A exactly one element of B.

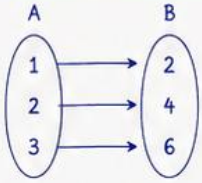
It is denoted by $f : A \to B$ or $f(x)$.

Note :
- Exactly one output for each input.
- Inputs may repeat, outputs may repeat.

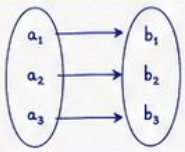
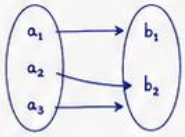
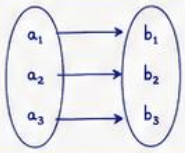
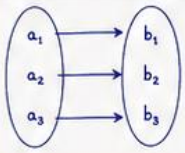
Function Notation

Domain	Set of all inputs (A)
Codomain	Set of all possible outputs (B)
Range (Image)	Set of actual outputs $f(A) \subseteq B$
Pre-image	For $b \in B$, $f^{-1}(b) = \{ a \in A \mid f(a) = b \}$

Ways to Represent a Function

1. Set of Ordered Pairs	2. Arrow Diagram	3. Mapping (Rule)
$f = \{(1,2), (2,4), (3,6)\}$ from $A = \{1,2,3\}$ to $B = \{2,4,6\}$		$f : N \to N$, $f(n) = 2n$

Types of Functions

Type	Definition	Diagram	Example
One-to-One (Injective)	Different inputs have different outputs. ($f(x_1) = f(x_2) \Rightarrow x_1 = x_2$)		$f : N \to N$, $f(n) = n + 1$
Many-to-One	Different inputs can have the same output.		$f : N \to N$, $f(n) = \lfloor n/2 \rfloor$
Onto (Surjective)	Every element of B is the image of at least one element of A. (Range = Codomain)		$f : R \to R$, $f(x) = x^3$
One-to-One Onto (Bijective)	Both one-to-one and onto.		$f : R \to R$, $f(x) = x + 2$

Domain, Codomain & Range

Let $f : A \to B$, $f = \{(1,2), (2,4), (3,4), (4,6)\}$

$A = \{1, 2, 3, 4\}$ (Domain)

$B = \{2, 4, 6, 8\}$ (Codomain)

$f(A) = \{2, 4, 6\}$ (Range / Image)

Example

Let $f : \mathbb{Z} \to \mathbb{Z}$ be defined by $f(x) = 3x - 1$

Find : (i) $f(2)$ (ii) $f(-1)$ (iii) $f(0)$

Sol : (i) $f(2) = 3(2) - 1 = 5$

(ii) $f(-1) = 3(-1) - 1 = -4$

(iii) $f(0) = 3(0) - 1 = -1$

Composition of Functions

If $f : A \to B$ and $g : B \to C$, then $g \circ f : A \to C$ is defined by

$(g \circ f)(x) = g(f(x))$ for all $x \in A$

Example : Let $f(x) = 2x + 1$ and $g(x) = x^2$

Then, $(g \circ f)(x) = g(f(x)) = g(2x + 1) = (2x + 1)^2$

and $(f \circ g)(x) = f(g(x)) = f(x^2) = 2x^2 + 1$

Inverse of a Function

A function $f : A \to B$ is invertible (has inverse) if it is bijective.

The inverse function is denoted by $f^{-1} : B \to A$ such that $f^{-1}(f(x)) = x$ for all $x \in A$ and $f(f^{-1}(y)) = y$ for all $y \in B$

Example : $f(x) = 2x + 3$ (from $R \to R$)

Let $y = 2x + 3 \Rightarrow x = \frac{y - 3}{2}$

So, $f^{-1}(y) = \frac{y - 3}{2}$

Restrictions

If $f : A \to B$ is a function and $C \subseteq A$, then the restriction of f to C is $f|_C : C \to B$ defined by

$(f|_C)(x) = f(x)$ for all $x \in C$

Example : $f(x) = x^2$, $A = R$

Let $C = \{-2, -1, 0, 1, 2\}$

Then, $f|_C = \{(-2,4), (-1,1), (0,0), (1,1), (2,4)\}$

Miscellaneous Facts

1. Domain of a function can be any set.
2. Range is always a subset of codomain.
3. Every one-to-one function has at most one inverse.
4. Identity function on A : $I_A(a) = a$ for all $a \in A$.
5. Constant function : $f(x) = c$ for all $x \in A$.

Quick Facts

- ☑ One-to-One $\Rightarrow$ Distinct inputs, distinct outputs.
- ☑ Onto $\Rightarrow$ All elements of codomain are hit.
- ☑ Bijective $\Rightarrow$ One-to-one and onto.
- ☑ Many-to-One $\Rightarrow$ At least two inputs have same output.
- ☑ Composition is associative : $h \circ (g \circ f) = (h \circ g) \circ f$
- ☑ Identity : $f \circ I_A = f$ and $I_B \circ f = f$

Think of a function as a machine that takes input from A and gives exactly one output in B.

6. Mathematical Induction

What is Mathematical Induction ?

Mathematical induction is a proof technique used to prove that a statement $P(n)$ is true for all natural numbers $n \geq n_0$ (usually $n_0 = 1$).

Note :

- Used for statements involving natural numbers.
- Works because every natural number is well-ordered.

Principle of Mathematical Induction

Let $P(n)$ be a statement defined for all integers $n \geq n_0$.

If the following two conditions hold:

1. Base Case	:	$P(n_0)$ is true.
2. Inductive Step	:	For any $k \geq n_0$, if $P(k)$ is true, then $P(k+1)$ is true.

Then $P(n)$ is true for all $n \geq n_0$.

Steps for Mathematical Induction

Step 1	Base Case Prove that $P(n_0)$ is true.
Step 2	Inductive Hypothesis Assume that $P(k)$ is true for some arbitrary $k \geq n_0$.
Step 3	Inductive Step Using the assumption in Step 2, prove that $P(k+1)$ is true.
Step 4	Conclusion By the principle of induction, $P(n)$ is true for all $n \geq n_0$.

Strong (Complete) Induction

If $P(n)$ is true for n_0, and for any $k > n_0$, if $P(n_0), P(n_0+1), \ldots, P(k)$ are true, then $P(k+1)$ is true, then $P(n)$ is true for all $n \geq n_0$.

Used when $P(k+1)$ depends on all previous cases n_0 to k (not just k).

Common Induction Patterns

① Linear Step	$P(k) \Rightarrow P(k+1)$	
② Quadratic Step	$P(k) \Rightarrow P(k+1)$	(expressions with squares)
③ Exponential Step	$P(k) \Rightarrow P(k+1)$	(powers of 2, 3, ...)
④ Divisibility	Prove property holds for $k+1$ using k	
⑤ Inequalities	Show LHS $\leq$ RHS for $k+1$ using k	

Common Mistakes

- × Skipping the base case.
- × Not using "assume $P(k)$ is true" in inductive step.
- × Proving $P(k+1) \Rightarrow P(k)$ (wrong direction).
- × Assuming $P(k+1)$ true without proof.
- × Not simplifying algebra correctly in inductive step.



★ Key Takeaway

Induction is like setting up dominoes.
If the first falls (base case) and each one knocks the next (inductive step),
then all will fall!

Example 1 : Sum of First n Natural Numbers

Prove that $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$ for all $n \geq 1$.

1. Base Case	$n = 1$ LHS $= 1$ $\frac{1(1+1)}{2} = 1$ ✓
2. Inductive Hypothesis	Assume $1 + 2 + 3 + \ldots + k = \frac{k(k+1)}{2}$
3. Inductive Step	For $k+1$: $1 + 2 + 3 + \ldots + k + (k+1)$ $= \frac{k(k+1)}{2} + (k+1)$ [Using I.H.] $= \frac{k^2 + k + 2k + 2}{2}$ $= \frac{(k+1)(k+2)}{2}$ ✓

Hence, true for all $n \geq 1$.

Example 2 : $2^n \geq n + 1$ for all $n \geq 1$

1. Base Case	$n = 1$ $\quad 2^2 = 2 \geq 2 = 1 + 1$ ✓
2. Inductive Hypothesis	Assume $2^k \geq k + 1$ for some $k \geq 1$
3. Inductive Step	For $k+1$: $2^{k+1} = 2 \cdot 2^k \geq 2(k+1)$ [Using I.H.] $= 2k + 2 \geq k + 2$ (since $2k + 2 \geq k + 2$ for $k \geq 1$) $= (k+1) + 1$ ✓

Hence, $2^n \geq n + 1$ for all $n \geq 1$.

Example 3 : (Strong Induction)

Prove that every integer $n \geq 2$ can be written as a product of primes.

1. Base Case	$n = 2$ $\quad 2$ is a prime. ✓
2. Inductive Hypothesis	Assume for all m with $2 \leq m \leq k$, each m can be written as a product of primes.
3. Inductive Step	Consider $k + 1$. If $k + 1$ is prime, done. If not, $k + 1 = a \cdot b$ where $2 \leq a, b \leq k$. By I.H., a and b are products of primes. So, $k + 1$ is also a product of primes. ✓

Hence, true for all $n \geq 2$.

7. Recurrence Relations

What is a Recurrence Relation ?

A recurrence relation is an equation that defines a sequence in terms of its preceding terms.

General form : $a_n = f(a_{n-1}, a_{n-2}, \cdots, a_{n-k})$ with given initial conditions.

Note :
- Used to model recursive processes.
- Common in algorithms and discrete math.

Components

Component	Description
Recurrence Relation	Equation defining a_n in terms of previous terms
Initial Conditions	Values for first few terms (e.g., a_0, a_1, ...)
Order	Number of previous terms used (highest index)
Homogeneous	a_n depends only on previous terms
Non-homogeneous	a_n includes an extra function $g(n)$

Order of Recurrence Relation

Order	Form	Example
1st Order	$a_n = f(a_{n-1})$	$a_n = 3a_{n-1}$, $a_0 = 2$
2nd Order	$a_n = f(a_{n-1}, a_{n-2})$	$a_n = a_{n-1} + a_{n-2}$, $a_0 = 0$, $a_1 = 1$
k^{th} Order	$a_n = f(a_{n-1}, \cdots, a_{n-k})$	$a_n = 2a_{n-1} - a_{n-2} + a_{n-3}$

Common Types of Recurrence Relations

Type	Form	Example
Linear Homogeneous (1st Order)	$a_n = c\,a_{n-1}$	$a_n = 2a_{n-1}$, $a_0 = 1$
Linear Homogeneous (2nd Order)	$a_n = p\,a_{n-1} + q\,a_{n-2}$	$a_n = a_{n-1} + a_{n-2}$, $a_0 = 0$, $a_1 = 1$
Linear Non-homogeneous (1st Order)	$a_n = c\,a_{n-1} + g(n)$	$a_n = a_{n-1} + 2$
Linear Non-homogeneous (2nd Order)	$a_n = p\,a_{n-1} + q\,a_{n-2} + g(n)$	$a_n = a_{n-1} + a_{n-2} + 1$

Example 1 : Fibonacci Sequence

Recurrence : $F_n = F_{n-1} + F_{n-2}$, $F_0 = 0$, $F_1 = 1$

n	0	1	2	3	4	6	7	8
F_n	0	1	1	3	5	8	13	21

Closed Form (Binet's Formula) :

$$F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}, \quad \text{where } \phi = \frac{1+\sqrt{5}}{2}, \ \psi = \frac{1-\sqrt{5}}{2}$$

Solving Techniques

① **Substitution Method**
Guess the pattern by computing initial terms and generalize.

② **Iteration (Expansion) Method**
Expand the relation repeatedly until a pattern emerges.

③ **Characteristic Equation Method**
Used for linear homogeneous recurrences with constant coefficients.

④ **Generating Functions**
Convert recurrence into algebraic equation using generating functions.

Example 2 : $a_n = 2a_{n-1} + 3$, $a_0 = 1$

1. Iteration	$a_1 = 2(1)+3 = 5$ $a_2 = 2(5)+3 = 13$ $a_3 = 2(13)+3 = 29$ Pattern looks like $a_n = 2^n \cdot 1 + 3(2^n - 1)$
2. General Form	$a_n = 2^n a_0 + 3(2^n - 1)$ $a_n = 2^n(1) + 3(2^n - 1) = 5 \cdot 2^n - 3$
3. Check	For $n = 0$: $a_0 = 5(1) - 3 = 2$ ✓ For $n = 1$: $a_1 = 5(2) - 3 = 7$ ✓ (Wait!?)

Oops! Let's re-check the pattern.

Correct general form:

$a_n = 2^n a_0 + 3(2^n - 1)$
$= 2^n + 3(2^n - 1) = 5 \cdot 2^n - 3$

Check $a_1 = 5(2) - 3 = 7$
$a_2 = 5(4) - 3 = 17$ ✓
Matches from iteration.

Example 3 : $a_n = a_{n-1} + a_{n-2} + 1$, $a_0 = 0$, $a_1 = 1$

Iteration :

$a_2 = 1 + 0 + 1 = 2$
$a_3 = 2 + 1 + 1 = 4$
$a_4 = 4 + 2 + 1 = 7$
$a_5 = 7 + 4 + 1 = 12$

Sequence : 0, 1, 2, 4, 7, 12, ...

This is Fibonacci-like with extra 1.

No simple closed form.

Applications

- ☑ Analysis of algorithms (e.g., Merge Sort, Tower of Hanoi)
- ☑ Counting problems (e.g., tilings, paths)
- ☑ Modeling recursive processes
- ☑ Dynamic programming

Key Takeaway

Understand the type of recurrence, solve using appropriate technique, and always use initial conditions to get the exact solution.

Common Recurrences (Quick Reference)

Recurrence	Closed Form
$a_n = c\,a_{n-1}$, $a_0 = a$	$a_n = a\,c^n$
$a_n = a_{n-1} + d$, $a_0 = a$	$a_n = a + nd$
$a_n = pa_{n-1} + q$, $a_0 = a$ $(p \neq 1)$	$a_n = p^n a + \frac{q(p^n - 1)}{p - 1}$
$a_n = a_{n-1} + a_{n-2}$, $a_0 = 0$, $a_1 = 1$	$a_n = F_n$ (Fibonacci numbers)

8. Counting Principles

What is Counting?

Counting is the process of determining the number of ways an event can occur.
Counting principles help us solve problems without listing all the possibilities.

Note :
- Order may or may not matter.
- Identify the rule first, then apply.

Addition Rule (Sum Rule)

If a task can be done in m ways or in n ways (but not both together), then total number of ways $= m + n$.

Example : A person can go to office either by bus (in 5 ways) or by train (in 3 ways).
Total ways $= 5 + 3 = 8$.

Multiplication Rule (Product Rule)

If a task consists of two independent steps, first can be done in m ways and second in n ways, then total number of ways $= m \times n$.

Example :
A password has 2 choices for first character, 3 choices for second and 4 choices for third.
Total passwords $= 2 \times 3 \times 4 = 24$.

Fundamental Principles

Principle	Description
Addition (Sum) Rule	If a task can be done in m ways or in n ways (mutually exclusive), then it can be done in $m + n$ ways.
Multiplication (Product) Rule	If a task consists of two steps, first can be done in m ways and second in n ways, then total ways $= m \times n$.
Complement Rule	If total possibilities $= N$ and event A can occur in n ways, then ways event A does not occur $= N - n$.

Generalized Product Rule

If a task consists of k independent steps and step i can be done in n_i ways (for $i = 1$ to k), then total number of ways

$$= n_1 \times n_2 \times \dots \times n_k = \prod_{i=1}^{k} n_i$$

Example :
A person has 3 shirts, 2 pants and 4 pairs of shoes.
Number of outfits $= 3 \times 2 \times 4 = 24$.

Common Counting Scenarios

Scenario	Rule Used	Formula	Example	Solution
Either A or B	Addition Rule	$m + n$	Choose a card from hearts (13) or from spades (13).	$13 + 13 = 26$
A then B (independent)	Multiplication Rule	$m \times n$	Choose a digit (10) then a letter (26) for a code.	$10 \times 26 = 260$
A or B or C (mutually exclusive)	Addition Rule	$m + n + p$	Choose one of red (4), blue (3) or green (5) pens.	$4 + 3 + 5 = 12$
A then B then C (independent)	Multiplication Rule	$m \times n \times p$	Choose a book (5), a pen (3) and a notebook (4).	$5 \times 3 \times 4 = 60$
None of A	Complement Rule	$N - n$	Total students $= 50$, who like tea $= 20$. Who do not like tea?	$50 - 20 = 30$

Special Cases

① At least one of the events occurs:
If events may overlap,
$|A \cup B| = |A| + |B| - |A \cap B|$

② Neither A nor B occurs:
Total $= N - |A \cup B|$

③ At least one of n events occurs:
$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum|A_i| - \sum|A_i \cap A_j| + \sum|A_i \cap A_j \cap A_k| - \dots$ (Inclusion-Exclusion)

Example :
In a class of 40 students, 18 like cricket, 15 like football and 8 like both.
How many like at least one of the games?
$|C \cup F| = 18 + 15 - 8 = 25$.

Example Problems

① A coin is tossed. If head, a die is rolled. If tail, a card is drawn from a deck.
Ways $= 1 \times 6$ (coin→die) $+ 1 \times 52$ (coin→card)
$= 6 + 52 = 58$.

② How many 3-digit numbers can be formed using digits 1, 2, 3, 4 without repetition?
Hundreds : 4 choices
Tens : 3 choices
Units : 2 choices
Total $= 4 \times 3 \times 2 = 24$.

③ A committee of 1 teacher and 2 students is to be selected from 5 teachers and 8 students.
Ways $= 5 \times C(8, 2) = 5 \times 28 = 140$.

Quick Facts

- ☑ Addition Rule ⇒ used for mutually exclusive choices.
- ☑ Multiplication Rule ⇒ used for independent sequential choices.
- ☑ Complement Rule ⇒ used for 'not this event' problems.
- ☑ Always identify whether order matters or not.
- ☑ Draw a tree diagram to visualize complex choices.
- ☑ Break the problem into steps.

Key Takeaway
Understand the situation, choose the correct principle, apply carefully, and count smartly!

9. Permutations & Combinations

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What are Permutations & Combinations?

Permutations deal with arrangements where order matters.

Combinations deal with selections where order does not matter.

Both help in counting the number of ways to arrange or choose items from a set.

Note :

- n = total number of distinct items
- r = number of items chosen ($r \leq n$)

Key Formulas

Concept	Formula
Factorial	$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$ $(0! = 1)$
Permutation (all r)	${}^nP_r = \frac{n!}{(n-r)!}$
Combination (all r)	${}^nC_r = \frac{n!}{r!(n-r)!}$
Relation	${}^nP_r = {}^nC_r \times r!$
Symmetry	${}^nC_r = {}^nC_{n-r}$
Sum of Combinations	$\sum_{r=0}^{n} {}^nC_r = 2^n$

Permutations (Order Matters)

Number of ways to arrange r items out of n distinct items in a particular order is ${}^nP_r = \frac{n!}{(n-r)!}$.

Example :

In how many ways can 3 students be arranged in a row from 8 students?

${}^8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$

Combinations (Order Does Not Matter)

Number of ways to choose r items from n distinct items is ${}^nC_r = \frac{n!}{r!(n-r)!}$.

Example :

In how many ways can 3 students be chosen from 8 students?

${}^8C_3 = \frac{8!}{3!5!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$

Permutations: Special Cases

Case	Formula	Example
All n items in a row	${}^nP_n = n!$	Arrange 5 books: $5! = 120$
With repetitions	$\frac{n!}{n_1!\, n_2! \dots, n_k!}$ ($n_1, n_2, \dots, n_k$ are repetitions)	Word: BALLOON Counts: B(1), A(1), L(2), O(2), N(1) Ways = $\frac{7!}{1!1!2!2!1!} = \frac{5040}{4} = 1260$
Circular arrangements	$(n-1)!$	5 people around a table: $(5-1)! = 24$
Circular (with repetitions)	$\frac{(n-1)!}{n_1!\, n_2! \dots, n_k!}$	6 people with 2 identical sisters: $\frac{(6-1)!}{2!} = 60$

Combinations: Special Cases

Case	Formula	Example
Choose none	${}^nC_0 = 1$	From any set, choose 0 item: 1 way
Choose all	${}^nC_n = 1$	From any set, choose all items: 1 way
Exactly r successes in n trials	${}^nC_r$	In 10 coin tosses, 3 heads: ${}^{10}C_3 = 120$
At most r successes in n trials	$\sum_{i=0}^{r} {}^nC_i$	At most 2 heads in 10 tosses: ${}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 = 1 + 10 + 45 = 56$

Relationship Between Permutations & Combinations

${}^nP_r = \frac{n!}{(n-r)!} = \frac{n!}{r!(n-r)!} \times r! = {}^nC_r \times r!$

Example :

${}^7P_4 = {}^7C_4 \times 4!$

$= 35 \times 24$

$= 840$

Key Takeaway

Use Permutations when order matters, Combinations when order does not matter.

Important Identities

1. ${}^nC_r = {}^nC_{n-r}$ (Symmetry)
2. ${}^nC_r = \frac{n}{r} \times {}^{n-1}C_{r-1}$ (Recursive)
3. ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ (Pascal's Rule)
4. $\sum_{r=0}^{n} {}^nC_r = 2^n$
5. $\sum_{r=0}^{n} r\,{}^nC_r = n2^{n-1}$

Pascal's Triangle (nC_r)

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
1   5  10  10   5   1
  ...
```

Example Problems

1. How many 4-digit numbers can be formed using digits 1, 2, 3, 4, 5 without repetition?

Solution :

${}^5P_4 = \frac{5!}{1!} = 120$

2. In how many ways can a committee of 3 be chosen from 10 people?

Solution :

${}^{10}C_3 = \frac{10!}{3!7!} = 120$

3. A word has letters: BOOKKEEPER. How many distinct arrangements are possible?

Solution :

Total letters = 10

Counts: B(2), O(2), K(2), E(3), P(1), R(1)

Ways = $\frac{10!}{2!2!2!3!1!1!} = 113400$

4. In how many ways can we choose at least 2 people from a group of 7?

Solution :

$\sum_{r=2}^{7} {}^7C_r = 2^7 - {}^7C_0 - {}^7C_1$

$= 128 - 1 - 7 = 120$

10. Pigeonhole Principle

What is Pigeonhole Principle ?

If n objects are placed into k boxes (pigeonholes), then at least one box must contain at least

$$\left\lceil \frac{n}{k} \right\rceil \text{ objects.}$$

Special case : If $n > k$, then at least one box contains at least 2 objects.

Note :

- n = number of objects
- k = number of boxes
- $\lceil x \rceil$ = the smallest integer greater than or equal to x.

Basic Examples

Situation	Conclusion
10 pigeons in 3 holes ($n = 10, k = 3$)	At least one hole has at least $\left\lceil \frac{10}{3} \right\rceil = 4$ pigeons.
7 apples in 4 baskets ($n = 7, k = 4$)	At least one basket has at least $\left\lceil \frac{7}{4} \right\rceil = 2$ apples.
13 students in 5 groups ($n = 13, k = 5$)	At least one group has at least $\left\lceil \frac{13}{5} \right\rceil = 3$ students.

Generalized Pigeonhole Principle

If n objects are placed into k boxes, then there exists at least one box containing at least $\left\lceil \frac{n}{k} \right\rceil$ objects.

Proof (by contradiction) :

Assume every box has at most $\lceil n/k \rceil - 1$ objects.

Then total objects $\leq k \times (\lceil \frac{n}{k} \rceil - 1) < n$

(since $k \times \lceil \frac{n}{k} \rceil - k < n$).

This contradicts the fact that we have n objects.

Hence, the statement is true.

Variations

Type	Statement
At least two in one box	If n objects are placed into k boxes and $n > k$, then at least one box contains at least 2 objects.
At least m in one box	If n objects are placed into k boxes and $n > k(m-1)$, then at least one box contains at least m objects.
Equal Pigeonhole Principle	If n objects are placed into n boxes, then at least one box has exactly one object.
Dirichlet's Box Principle	Same as Pigeonhole Principle, commonly used in number theory.

Common Applications

Application	How to Apply	Example
Divisibility	Consider remainders when dividing by k.	Among any 7 integers, at least two have the same remainder when divided by 6. ($n = 7, k = 6 \Rightarrow$ at least 2 same remainder)
Dates / Months	12 months → 12 pigeonholes.	In any group of 13 people, at least two have birthdays in the same month. ($n = 13, k = 12 \Rightarrow$ at least 2 same month)
Colors / Categories	Objects colored / categorized into limited groups.	If 5 points are plotted in a plane, at least 3 are collinear. (Pigeonholes are the lines determined by any two points.)
Numbers / Remainders	Use remainders modulo k.	In any set of 20 integers, two numbers have the same remainder when divided by 19. ($n = 20, k = 19$)

Example Problems

① Show that among any 11 people, at least two were born in the same month.

Solution :

$n = 11$ people, $k = 12$ months

At least one month has at least $\left\lceil \frac{11}{12} \right\rceil = 1 + 1 = 2$ people.

Hence, at least two have the same birth month.

② Prove that in any group of 25 people, at least 3 were born on the same day of the week.

Solution :

There are 7 days in a week.

$n = 25, k = 7$

At least one day has at least $\left\lceil \frac{25}{7} \right\rceil = 4$ people.

So, at least 3 people share the same day.

③ Among any 367 integers, prove that at least two have the same remainder when divided by 366.

Solution :

Remainders can be 0, 1, 2, ..., 365

$\Rightarrow k = 366$

$n = 367 > 366$

Hence, at least two integers have the same remainder.

Key Takeaway

- ☑ Powerful counting tool.
- ☑ Helps prove existence without construction.
- ☑ Works in many areas: number theory, geometry, combinatorics, algorithms.
- ☑ When in doubt, think pigeonhole!

11. Probability Basics

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What is Probability?

Probability is a measure of how likely an event is to occur.

It ranges from 0 (impossible) to 1 (certain).

$$0 \le P(E) \le 1$$

Note :

- S = Sample Space
- E = Event
- n(S) = Total outcomes
- n(E) = Favorable outcomes

Probability Axioms

No.	Axiom	Description
1.	Non-negativity	$P(E) \ge 0$ for any event E
2.	Normalization	$P(S) = 1$
3.	Additivity	If $E_1, E_2, \ldots, E_n$ are pairwise disjoint events, then $P(E_1 \cup E_2 \cup \ldots \cup E_n) = \sum_{i=1}^{n} P(E_i)$

Classical (Theoretical) Probability

If all outcomes in S are equally likely,

$$P(E) = \frac{n(E)}{n(S)} = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

Example :

A fair die is rolled. What is the probability of getting an even number?

$S = \{1,2,3,4,5,6\}$, $n(S) = 6$

$E = \{2,4,6\}$, $n(E) = 3$

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

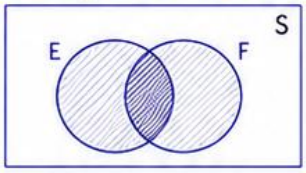
Types of Events

Type	Description
Simple / Elementary	Event with exactly one outcome.
Compound	Event with two or more outcomes.
Certain Event	Event that always occurs. $P(E) = 1$
Impossible Event	Event that never occurs. $P(E) = 0$
Complementary	Event $\bar{E}$ $\bar{E}$ = "not E" $P(\bar{E}) = 1 - P(E)$

Common Formulas

1. Complement Rule
   $P(\bar{E}) = 1 - P(E)$
2. Addition Rule (General)
   $P(E \cup F) = P(E) + P(F) - P(E \cap F)$
3. Addition Rule (Mutually Exclusive)
   If $E \cap F = \phi$,
   $P(E \cup F) = P(E) + P(F)$
4. Conditional Probability
   $P(E \mid F) = \frac{P(E \cap F)}{P(F)}$, $P(F) > 0$
5. Multiplication Rule
   $P(E \cap F) = P(E) \times P(F \mid E)$
   $= P(F) \times P(E \mid F)$

Venn Diagram for Two Events



- E only = E − F
- F only = F − E
- Both = $E \cap F$
- Neither = $(E \cup F)'$

Conditional Probability

Probability of E occurring given that F has occurred.

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)}, \quad P(F) > 0$$

Example :

From a deck of 52 cards, one card is drawn. What is the probability that it is a King given that it is a Heart?

$P(\text{King} \cap \text{Heart}) = 1$ (only King of Hearts)

$P(\text{Heart}) = 13$

$$P(\text{King} \mid \text{Heart}) = \frac{1}{13}$$

Independent Events

Events E and F are independent if

$$P(E \cap F) = P(E) \times P(F)$$

(occurrence of one does not affect the other)

Example :

A coin is tossed and a die is rolled.

Let E = getting Head, $P(E) = \frac{1}{2}$

Let F = getting a 3, $P(F) = \frac{1}{6}$

$$P(E \cap F) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12} = P(E) \times P(F)$$

Hence, E and F are independent.

Important Properties

1. $0 \le P(E) \le 1$
2. $P(S) = 1$
3. $P(\phi) = 0$
4. $P(\bar{E}) = 1 - P(E)$
5. If $E \le F$, then $P(E) \le P(F)$

Key Takeaway

Probability helps us quantify uncertainty. Master the basics, and complex problems become much easier!

Quick Examples

Situation	Calculation	Probability
Rolling a 4 on a fair die	$n(E) = 1$, $n(S) = 6$	$P(E) = \frac{1}{6}$
Getting a red card from a deck	$n(E) = 26$, $n(S) = 52$	$P(E) = \frac{26}{52} = \frac{1}{2}$
Getting head on a fair coin	$n(E) = 1$, $n(S) = 2$	$P(E) = \frac{1}{2}$
Getting prime number on a die (2,3,5)	$n(E) = 3$, $n(S) = 6$	$P(E) = \frac{3}{6} = \frac{1}{2}$

Practice Problems

1. A coin is tossed twice. What is the probability of getting exactly one Head?

Solution :

Sample space = {HH, HT, TH, TT}

Favorable = {HT, TH}

$$P = \frac{2}{4} = \frac{1}{2}$$

2. A bag has 3 red, 4 blue and 5 green balls. One ball is drawn. What is the probability that it is blue?

Solution :

Total = 3 + 4 + 5 = 12

Favorable = 4

$$P = \frac{4}{12} = \frac{1}{3}$$

3. Two events E and F are such that $P(E) = 0.4$, $P(F) = 0.5$ and $P(E \cap F) = 0.2$. Find $P(E \cup F)$.

Solution :

$P(E \cup F) = P(E) + P(F) - P(E \cap F)$

$= 0.4 + 0.5 - 0.2$

$= 0.7$

4. A card is drawn from a deck. What is the probability that it is (i) a King (ii) not a King?

Solution :

(i) $P(\text{King}) = \frac{4}{52} = \frac{1}{13}$

(ii) $P(\text{not King}) = 1 - \frac{1}{13} = \frac{12}{13}$

12. Graph Theory Fundamentals

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What is Graph Theory ?

Graph Theory is the study of graphs, which are mathematical structures used to model pairwise relationships between objects.

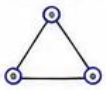
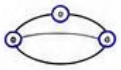
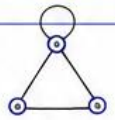
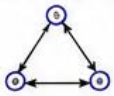
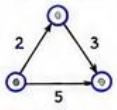
A graph G is denoted as
$G = (V, E)$
where V = set of vertices (nodes) and E = set of edges.

Note :
- Simple graph → no loops and no multiple edges
- Undirected graph → edges have no direction
- Directed graph (digraph) → edges have direction

Basic Terminology

Term	Definition
Vertex (or Node)	A point in the graph.
Edge	A line connecting two vertices.
Undirected Graph	Edges have no direction.
Directed Graph (Digraph)	Edges have a direction (u → v).
Adjacent Vertices	Two vertices connected by an edge.
Incident Edge	An edge is incident on a vertex if it is connected to it.
Degree deg(v)	Number of edges incident on v (in digraphs → in-degree, out-degree).
Path	A sequence of vertices where each consecutive pair is connected.
Cycle	A path that starts and ends at the same vertex (length ≥ 3).
Connected Graph	There is a path between every pair of vertices.

Types of Graphs

Simple Graph	Multigraph	Pseudograph	Directed Graph	Weighted Graph
				
No loops, no multiple edges	Multiple edges allowed	Loops and multiple edges allowed	Edges have direction	Edges have weights/costs

Handshaking Lemma

In an undirected graph G = (V, E), the sum of degrees of all vertices equals twice the number of edges.

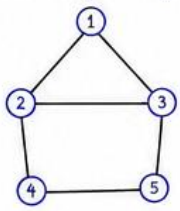
$$\sum_{v \in V} \deg(v) = 2|E|$$

Example :
From the example graph,
Sum of degrees = 2 + 3 + 3 + 2 + 2 = 12
Number of edges = 6
$2|E| = 2 \times 6 = 12$ ✓

Degree in Graphs

Graph Type	Degree of Vertex v
Undirected Graph	deg(v) = number of edges incident on v (A loop contributes 2 to degree)
Directed Graph (Digraph)	In-degree $\deg^-(v)$ = number of edges coming into v Out-degree $\deg^+(v)$ = number of edges going out of v Total degree $\deg(v) = \deg^-(v) + \deg^+(v)$

Example Graph



$V = \{1, 2, 3, 4, 5\}$
$E = \{(1,2), (1,3), (2,3), (2,4), (3,5), (4,5)\}$

$\deg(1) = 2$
$\deg(2) = 3$
$\deg(3) = 3$
$\deg(4) = 2$
$\deg(5) = 2$

Important Concepts

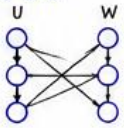
① **Path**
A path from u to v is a sequence $u = v_0, v_1, v_2, \ldots, v_k = v$ such that each $(v_i, v_{i+1}) \in E$.

② **Cycle**
A path with $v_0 = v_k$ and $k \geq 3$, with all other vertices distinct.

③ **Connected Components**
Maximal connected subgraphs.

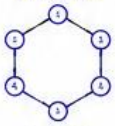
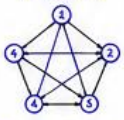
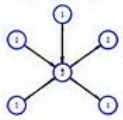
④ **Complete Graph K_n**
Every pair of distinct vertices is connected.
Number of edges = $\frac{n(n-1)}{2}$

⑤ **Bipartite Graph**
Vertices can be divided into two disjoint sets U and W such that every edge connects a vertex from U to W.



⑥ **Tree**
A connected acyclic undirected graph. For n vertices, number of edges = n − 1.

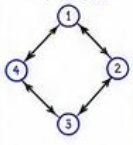
Common Graphs

Path Graph P_n (n vertices)	Cycle Graph C_n (n vertices)	Complete Graph K_n (n vertices)	Star Graph S_n (n vertices)
			
n − 1 edges	n edges	$n(\frac{n-1}{2}) - 2$ edges	n − 1 edges

Directed Graph Concepts

Term	Definition
In-degree $\deg^-(v)$	Number of edges directed into v
Out-degree $\deg^+(v)$	Number of edges directed out of v
Source	Vertex with in-degree 0
Sink	Vertex with out-degree 0
Strongly Connected (Digraph)	Every vertex is reachable from every other vertex

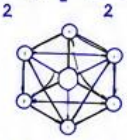
Example Digraph



$\deg^-(1) = 1, \deg^+(1) = 1$
$\deg^-(2) = 1, \deg^+(2) = 1$
$\deg^-(3) = 1, \deg^+(3) = 1$
$\deg^-(4) = 1, \deg^+(4) = 1$

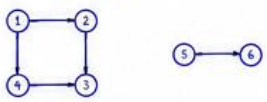
Example Problems

① How many edges are there in K_6 ?
Solution :
$|E| = \frac{6(6-1)}{2} = \frac{6 \times 5}{2} = 15$



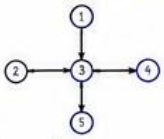
② In a graph with 7 vertices, the sum of degrees is 18. How many edges does the graph have ?
Solution :
$2|E| = 18 \Rightarrow |E| = 9$

③ Is the following graph connected?



Solution : No
(Two connected components)

④ Is the following graph a tree?



Solution : Yes
(Connected and acyclic, n = 5, |E| = 4 = n − 1)

Key Takeaway

Graphs help model real-world relationships. Understanding basic terms and properties is the foundation for advanced topics like algorithms and network design.

Applications

- ☑ Social Networks
- ☑ Computer Networks
- ☑ Maps & Navigation
- ☑ Scheduling & Dependencies
- ☑ Circuit Design
- ☑ Web Structure (SEO)
- ☑ Recommendation Systems

Quick Facts

- In a tree with n vertices → edges = n − 1
- In a connected graph with n vertices → minimum edges = n − 1, maximum edges = $\frac{n(n-1)}{2}$
- Every undirected graph with odd degree vertices is even.

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13. Graph Types & Representations

What are Graph Types & Representations ?

Graphs can be classified into different types based on edges and direction. To store graphs in computers, we use representations like Adjacency Matrix and Adjacency List.

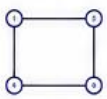
Note :
- Graph = (V, E)
- V = set of vertices
- E = set of edges
- |V| = number of vertices (n)
- |E| = number of edges (m)

Quick Comparison

Type	Direction	Multiple Edges	Self Loop
Undirected	No	No	No
Multigraph	No	Yes	Yes
Directed (Digraph)	Yes	No	No
Directed Multigraph	Yes	Yes	Yes

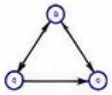
Types of Graphs

Undirected Graph



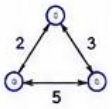
Edges have no direction.

Directed Graph (Digraph)



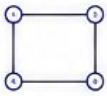
Edges have direction.

Weighted Graph



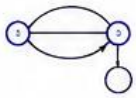
Edges have weights or costs.

Unweighted Graph



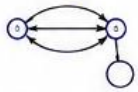
All edges have equal weight.

Multigraph



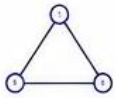
Multiple edges between same pair & loops allowed.

Directed Multigraph



Multiple directed edges & loops allowed.

Simple Graph



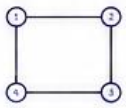
No loops and no multiple edges.

Graph Representations

1. Adjacency Matrix

- A 2D matrix of size n × n.
- If edge exists between i and j, matrix[i][j] = 1 (or weight).
- Otherwise 0 (or ∞ for weighted graphs).
- For undirected graph, matrix is symmetric.

Example (Undirected Graph)



Adjacency Matrix

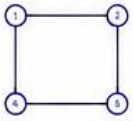
	1	2	3	4
1	0	1	0	1
2	1	0	1	0
3	0	1	0	1
4	1	0	1	0

Characteristics
- Space Complexity: $O(n^2)$
- Edge existence check: O(1)
- Good for dense graphs.

2. Adjacency List

- For each vertex, store a list of its adjacent vertices.
- Use array/vector of lists.
- For weighted graph, store (neighbor, weight).

Example (Undirected Graph)



Adjacency List

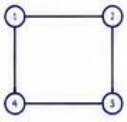
1 → 2, 4
2 → 1, 3
3 → 2, 4
4 → 1, 3

Characteristics
- Space Complexity: O(n + m)
- Edge existence check: O(deg(v))
- Good for sparse graphs.

3. Edge List

- Store all edges as a list of pairs (u, v) or (u, v, w) for weighted graphs.

Example (Undirected Graph)



Edge List

(1, 2)
(1, 4)
(2, 3)
(3, 4)

Characteristics
- Space Complexity: O(m)
- Simple to implement.
- Useful for algorithms like Kruskal's MST.

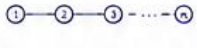
Special Graphs

Complete Graph (K_n)



Every pair of distinct vertices is connected.

Path Graph (P_n)



A simple path of length n-1.

Cycle Graph (C_n)



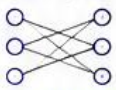
A closed path of length n.

Star Graph (S_n)



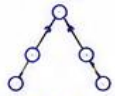
One center connected to all others.

Bipartite Graph ($K_{m,n}$)



Vertices divided into two sets; edges only across sets.

Tree



Connected, acyclic, |E| = |V| - 1

Disjoint Graph



Two or more components with no edges between them.

Null Graph (N_n)



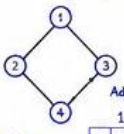
No edges at all.

Representations Comparison

Feature	Adjacency Matrix	Adjacency List	Edge List
Space	$O(n^2)$	O(n + m)	O(m)
Best For	Dense Graphs	Sparse Graphs	Algorithms (Kruskal, etc.)
Edge Check (u, v)	O(1)	O(deg(u))	O(m)
Iterate Neighbors of u	O(n)	O(deg(u))	O(m)
Add Edge	O(1)	O(1)	O(1)
Remove Edge	O(1)	O(deg(u))	O(m)

Example Problems

① Given the graph below, write its adjacency matrix and adjacency list.



Solution :

Adjacency Matrix

	1	2	3	4
1	0	1	1	1
2	1	0	1	0
3	1	1	0	1
4	1	0	1	0

Adjacency List

1 → 2, 3, 4
2 → 1, 3
3 → 1, 2, 4
4 → 1, 3

② How many edges are there in a comple with 6 vertices?

Solution :

In complete graph K_n,

$$|E| = \frac{n(n-1)}{2}$$

For n = 6,

$$|E| = \frac{6 \times 5}{2} = 15$$

③ Represent the following edges as an edge list.

Edges = {(1,2),(1,3),(2,4),(3,4),(4,5)}

Solution :

Edge List

(1, 2)
(1, 3)
(2, 4)
(3, 4)
(4, 5)

④ For the graph with n = 1000 vertices and m = 2000 edges, which representation is most space efficient?

Solution :

Adjacency Matrix → $O(n^2) = 10^6$ (large)
Adjacency List → O(n + m) = 3000 ✓
Edge List → O(m) = 2000 (best)
Edge list is most space efficient.

Key Takeaway

Choose representation based on the graph density and operation you need. Understand graph types well, as they form the base for advanced algorithms!

When to Use

- ☑ Adjacency Matrix → When graph is dense
- ☑ Adjacency List → When graph is sparse
- ☑ Edge List → For algorithms & simple storage

Remember

- ☑ Simple graph: no loops, no multiple edges
- ☑ Tree is a special kind of connected acyclic graph
- ☑ |E| = n - 1 for a tree

14. Trees & Spanning Trees

codedsoul_05

What is a Tree ?

A tree is a connected, acyclic graph.
It has n vertices and n - 1 edges.
There is exactly one simple path between any two vertices.

Tree = Connected + Acyclic
Edges = n - 1

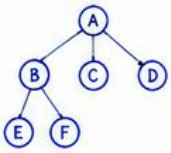
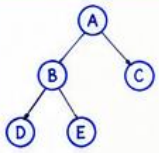
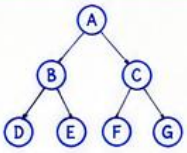
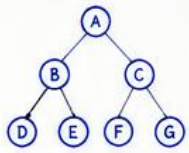
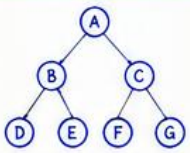
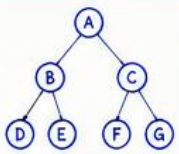
Note :

- If a graph is connected and has n - 1 edges, then it is a tree.
- If a graph has a cycle, it is not a tree.

Basic Terminology

Term	Definition	Example
Root	The top node (in rooted trees).	A
Parent	A node with direct children.	A is parent of B, C
Child	A node below a parent.	B, C are children of A
Leaf	A node with no children.	D, E, F
Internal node	A node with at least one child.	A, B, C
Degree	Number of children of a node.	A has degree 2

Types of Trees

General Tree	Binary Tree	Full Binary Tree	Complete Binary Tree	Perfect Binary Tree	Balanced Binary Tree
					
No restrictions on number of children.	Each node has at most 2 children.	Every internal node has exactly 2 children.	All levels are filled except possibly the last, which is filled from left.	All internal nodes have 2 children and all leaves are at same level.	Height difference at any node ≤ 1.

Properties of Trees

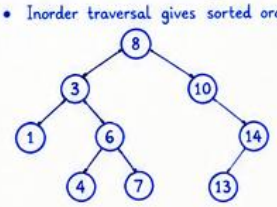
1. For a tree with n vertices, number of edges = n - 1.
2. Sum of degrees of all vertices = 2(n - 1).
3. A tree with n vertices has n - 1 edges.
4. A tree with n vertices has exactly n - 1 leaves (in a full binary tree, leaves = $\lceil \frac{n}{2} \rceil$).
5. A tree is connected and acyclic.

Tree Traversals

Traversal	Order	Use
Preorder (DFS)	Root → Left → Right	Copying, expression trees
Inorder (DFS)	Left → Root → Right	Sorted order (BST)
Postorder (DFS)	Left → Right → Root	Deletion, expression trees
Level Order (BFS)	Level by level	Shortest path, level info

Binary Search Tree (BST)

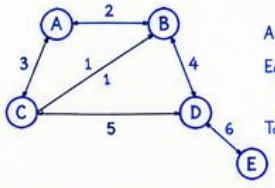
- Left subtree < root < right subtree.
- Inorder traversal gives sorted order.



Minimum Spanning Tree (MST)

A spanning tree of a connected, weighted graph with the minimum possible total edge weight.

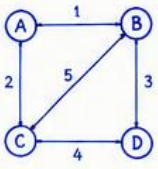
It has: • n - 1 edges • connected • no cycles



A possible MST:
Edges: (C, B) = 1, (A, B) = 2, (B, D) = 4, (D, E) = 6
Total weight = 13

Spanning Tree

A spanning tree of a graph is a tree that contains all the vertices of the graph.
It may not be the minimum weight.



Possible Spanning Tree:
Edges: (A, B), (A, C), (C, D)
Total edges = 3
(n = 4 → n - 1 = 3)

Key Differences

Feature	Tree	Spanning Tree
Definition	Connected, acyclic	Subgraph of a graph with all vertices, no cycles
Edges	n - 1	n - 1
Vertices	n	Same as original graph
Cycle	No	No
Weight	Not required	Usually used for minimum weight

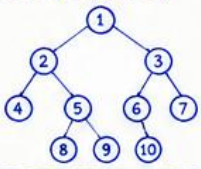
Algorithms for MST

Kruskal's Algorithm	Prim's Algorithm
1. Sort edges by weight.	1. Start from any vertex.
2. Pick the smallest edge that doesn't create a cycle.	2. Add the minimum weight edge that connects to a new vertex.
3. Repeat until you have n - 1 edges.	3. Repeat until all vertices are included.
Time Complexity : O(E log E)	Time Complexity : O(E log V)

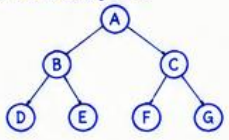
Example Problems

1. A tree has 10 vertices. How many edges does it have?

Solution :
Edges = n - 1 = 10 - 1 = 9



2. What is the degree of each vertex in the following tree?



A = 2, B = 3, C = 3,
D = 0, E = 0, F = 0, G = 0

Key Takeaways

- ☑ Tree = connected + acyclic
- ☑ Edges in a tree = n - 1
- ☑ Spanning tree includes all vertices
- ☑ MST has minimum total weight
- ☑ Kruskal = sort + union-find
- ☑ Prim = grow from a start node

Useful Formulas

- Edges in tree : n - 1
- Sum of degrees : 2(n - 1)
- Leaves (full binary) : $\lceil \frac{n}{2} \rceil$

Remember

- Tree ≠ Spanning Tree
- All trees are spanning trees
- Not all spanning trees are minimum
- MST is a special spanning tree

15. Graph Algorithms

What are Graph Algorithms?

Graph algorithms are used to solve problems on graphs by traversing, searching, finding paths, connecting components, etc.

Note :

- V = set of vertices
- E = set of edges
- |V| = number of vertices (n)
- |E| = number of edges (m)

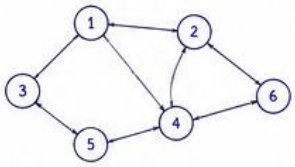
Common Applications

- ☑ V = set path (Google Maps)
- ☑ Network routing
- ☑ Social networks
- ☑ Dependency resolution
- ☑ Cycle detection
- ☑ Minimum cost networks
- ☑ Scheduling & precedence
- ☑ And many more...

Graph Representation Used

- Adjacency List (preferred for most algorithms)
- Works for both directed and undirected graphs.

Example Graph (Undirected)



Adjacency List

1: 2, 3
2: 1, 4
3: 1, 4, 5
4: 2, 3, 5, 6
5: 3, 4
6: 4

— Common Graph Algorithms —

1. BFS (Breadth First Search)

Visits nodes level by level using a queue.

Use Cases
- Shortest path in unweighted graph
- Level order traversal

Time Complexity: O(V + E)
Space Complexity: O(V)

Pseudo Code (DFS)
1. Create a queue Q
2. Mark start node visited and push to Q
3. While Q not empty:
   a. Pop front node u
   b. Visit u
   c. Push all unvisited neighbors of u

2. DFS (Depth First Search)

Explores as far as possible along a path using recursion or stack.

Use Cases
- Connected components
- Cycle detection
- Topological sort

Time Complexity: O(V + E)
Space Complexity: O(V)

Pseudo Code (DFS)
1. Mark start node visited
2. For each unvisited neighbor v: DFS(v)

3. Dijkstra's Algorithm (Shortest Path)

Finds shortest path from a source to all other vertices (works for non-negative weights).

Use Cases
- Shortest path in weighted graph
- Navigation systems

Time Complexity: O((V + E) log V) (Using Min-Heap)

Steps
1. Initialize dist[src] = 0, others = ∞
2. Push (0, src) to min-heap
3. While heap not empty:
   a. Pop node u with min dist
   b. For each neighbor v of u:
      if dist[v] > dist[v] + w(u,v): update dist[v] and push

4. Bellman-Ford Algorithm (Shortest Path)

Finds shortest path even with negative weights.
Detects negative weight cycle.

Use Cases
- Shortest path in graph with negative edges

Time Complexity: O(V × E)

Steps
1. Initialize dist[src] = 0, others = ∞
2. Relax all edges (V - 1) times
3. Check for negative cycle by relaxing once more

5. Floyd-Warshall Algorithm (All Pairs Shortest Path)

Finds shortest paths between all pairs of vertices.

Use Cases
- All pairs shortest path
- Dense graphs

Time Complexity: O(V³)
Space Complexity: O(V²)

Steps
1. Create dist matrix
2. For each intermediate k
3. For each i, j:
   dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j])

6. Topological Sort (For DAG)

Linear ordering of vertices in a Directed Acyclic Graph (DAG).

Use Cases
- Task scheduling
- Dependency resolution

Time Complexity: O(V + E)

Method (Kahn's Algorithm)
1. Compute in-degree of all nodes
2. Push all nodes with in-degree 0 into queue
3. While queue not empty:
   a. Pop u, add to result
   b. Reduce in-degree of neighbors of u
   c. If in-degree becomes 0, push it

7. Kruskal's Algorithm (MST)

Finds Minimum Spanning Tree using Disjoint Set (Union-Find).

Use Cases
- Minimum cost to connect all vertices

Time Complexity: O(E log E)

Steps
1. Start all edges by weight
2. Make each vertex its own set
3. For each edge (u, v):
   If find(u) ≠ find(v):
   Add edge to MST
   Union(u, v)

8. Prim's Algorithm (MST)

Finds Minimum Spanning Tree by growing from a start vertex.

Use Cases
- Minimum cost to connect all vertices

Time Complexity: O(E log V)

Steps
1. Start from any vertex
2. Push its edges to min-heap
3. While heap not empty:
   a. Pop min edge (u, v)
   b. If v not visited:
      Add to MST and push edges of v

9. Cycle Detection

Undirected Graph
- Use DFS/BFS
- If we visit an already visited node that is not parent → cycle

Directed Graph
- Use DFS with recursion stack
- If we visit a node already in recursion stack → cycle

10. Strongly Connected Components (SCC)

Finds SCC in a directed graph.

Use Cases
- Code dependencies
- Web link analysis

Algorithms
- Kosaraju's Algorithm (O(V+E))
- Tarjan's Algorithm (O(V+E))

11. Articulation Points (Cut Vertices)

Removing these vertices increases number of connected components.

Algorithm: DFS based
Time Complexity: O(V + E)

12. Bridges (Cut Edges)

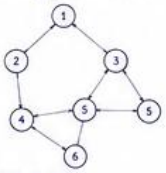
Removing these edges increases number of connected components.

Algorithm: DFS based
Time Complexity: O(V + E)

— Example Problems —

7. BFS Traversal

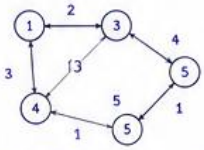
Given graph, start from 1.



BFS Order:
1, 2, 3, 4, 5, 6

2. Dijkstra (Shortest Path)

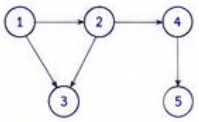
Find shortest path from 1 to 5.



Shortest Path: 1 → 2 → 5
Total Cost = 5

3. Topological Sort

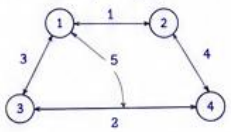
Given DAG, find topological order.



One Possible Order:
1, 2, 3, 4, 5

4. MST (Kruskal)

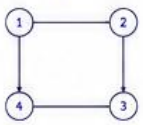
Find MST of the graph.



MST Edges: (1-2,1), (3-4,2), (1-3,3)
Total Weight = 1 + 2 + 3 = 6

5. Cycle Detection (Undirected)

Does cycle exist?



Yes, cycle exists.

6. Bellman-Ford

Edges: (1,2,4), (1,3,5), (2,3,-2), (2,4,6), (3,4,1)
Find shortest path from 1.

Shortest Distances:
1: 0
2: 4
3: 2 (1 → 2 → 3)
4: 3 (1 → 2 → 3 → 4)

Key Takeaway

Different algorithms are used for different problems. Understanding their use cases and complexities helps in applying the right tool efficiently!

Complexity Quick Reference

Algorithm	Time Complexity
BFS / DFS	O(V + E)
Dijkstra (Min-Heap)	O((V + E) log V)
Bellman-Ford	O(V × E)
Floyd-Warshall	O(V³)
Topological Sort	O(V + E)
Kruskal (MST)	O(E log E)
Prim (MST)	O(E log V)

When to Use

- BFS → shortest path in unweighted graph
- DFS → traversal, cycle detection, SCC
- Dijkstra → shortest path (no negative weight)
- Bellman-Ford → shortest path (with negative weight)
- Floyd-Warshall → all pairs shortest path
- Topological Sort → DAG ordering
- Kruskal / Prim → Minimum Spanning Tree

Remember

- ☑ Choose algorithm based on graph type and problem requirement.
- ☑ Understand time complexity.
- ☑ Practice on different examples.
- ☑ Implementation + intuition = Mastery!

16. Boolean Algebra & Logic Gates

What is Boolean Algebra ?

Boolean Algebra is a branch of algebra that works with binary variables (0 or 1) using operations like AND, OR, NOT.

Note :

- 0 = False, OFF, Low
- 1 = True, ON, High
- + = OR
- · = AND
- ′ = NOT (Complement)

Laws of Boolean Algebra

Law	Name	AND (·)
①	Commutative	$A + B = B + A$	$A \cdot B = B \cdot A$
②	Associative	$A + (B + C) = (A + B) + C$	$A \cdot (B \cdot C) = (A \cdot B) \cdot C$
③	Distributive	$A \cdot (B + C) = A \cdot B + A \cdot C$	$A + (B \cdot C) = (A + B) \cdot (A + C)$
④	Identity	$A + 0 = A$	$A \cdot 1 = A$
⑤	Null / Dominance	$A + 1 = 1$	$A \cdot 0 = 0$
⑥	Idempotent	$A + A = A$	$A \cdot A = A$
⑦	Complement	$A + A' = 1$	$A \cdot A' = 0$
⑧	Involution (Double Negation)	$(A')' = A$	$(A)' = A$
⑨	Absorption	$A + A \cdot B = A$	$A \cdot (A + B) = A$
⑩	De Morgan's Theorems	$(A + B)' = A' \cdot B'$	$(A \cdot B)' = A' + B'$

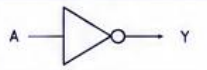
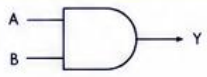
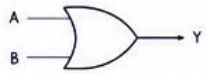
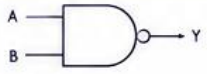
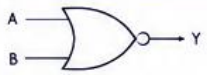
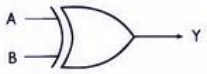
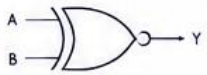
Basic Identities

1. $A + 0 = A$
2. $A + 1 = 1$
3. $A \cdot 0 = 0$
4. $A \cdot 1 = A$
5. $A + A = A$
6. $A \cdot A = A$
7. $A + A' = 1$
8. $A \cdot A' = 0$
9. $(A')' = A$

Properties

- Closure: Result is always 0 or 1.
- Commutative: Order does not matter.
- Associative: Grouping does not matter.
- Distributive: AND over OR and OR over AND.
- Complementarity: Every variable has a complement.

Logic Gates

Gate	Symbol	Boolean Expression	Truth Table (A, B → Y)	Description
1. NOT (Inverter)		$Y = A'$	A: 0 1 / Y: 1 0	Outputs the complement of the input.
2. AND		$Y = A \cdot B$	A B Y: 0 0 0; 0 1 0; 1 0 0; 1 1 1	Outputs 1 only when all inputs are 1.
3. OR		$Y = A + B$	A B Y: 0 0 0; 0 1 1; 1 0 1; 1 1 1	Outputs 1 when at least one input is 1.
4. NAND		$Y = (A \cdot B)'$	A B Y: 0 0 1; 0 1 1; 1 0 1; 1 1 0	NOT of AND.
5. NOR		$Y = (A + B)'$	A B Y: 0 0 1; 0 1 0; 1 0 0; 1 1 0	NOT of OR.
6. XOR (Exclusive OR)		$Y = A \oplus B$ $= A' \cdot B + A \cdot B'$	A B Y: 0 0 0; 0 1 1; 1 0 1; 1 1 0	Outputs 1 when inputs are different.
7. XNOR (Exclusive NOR)		$Y = (A \oplus B)'$ $= A \cdot B + A' \cdot B'$	A B Y: 0 0 1; 0 1 0; 1 0 0; 1 1 1	Outputs 1 when inputs are same.

Important Theorems

- De Morgan's Theorems:
  $(A + B)' = A' \cdot B'$
  $(A \cdot B)' = A' + B'$
- Double Negation: $(A')' = A$
- Idempotent Laws:
  $A + A = A$, $A \cdot A = A$
- Absorption Laws:
  $A + A \cdot B = A$
  $A \cdot (A + B) = A$
- Consensus Theorem:
  $A \cdot B + A' \cdot C + B \cdot C = A \cdot B + A' \cdot C$

Boolean Expression Precedence (High → Low)

1. NOT (′)
2. AND (·)
3. OR (+)

(Use Parentheses when in doubt)

Universal Gates

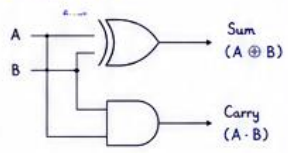
- NAND gate can be used to implement any logic circuit.
- NOR gate can also be used to implement any logic circuit.

Useful Identities

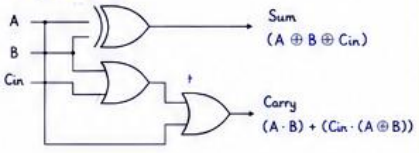
1. $A + 0 = A$
2. $A + 1 = 1$
3. $A \cdot 0 = 0$
4. $A \cdot 1 = A$
5. $A + A' = 1$
6. $A \cdot A' = 0$
7. $(A + B)' = A' \cdot B'$
8. $(A \cdot B)' = A' + B'$

Combinational Circuits Examples

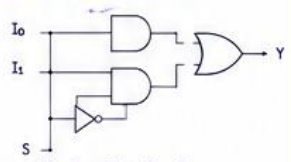
1. Half Adder



2. Full Adder

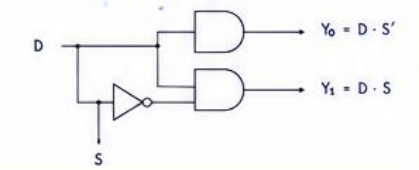


3. Multiplexer (2:1 MUX)



$Y = I_0 \cdot S' + I_1 \cdot S$

4. Demultiplexer (1:2 DEMUX)

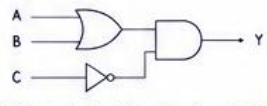


Example Problems

① Simplify using Boolean Algebra:
$Y = A \cdot B + A \cdot B' + A' \cdot B$

Solution:
$Y = A \cdot (B + B') + A' \cdot B$
$Y = A \cdot 1 + A' \cdot B = A + A' \cdot B$
$Y = (A + A') (A + B) = 1 \cdot (A + B)$
$Y = A + B$

② Implement $Y = (A + B) \cdot C'$ using logic gates.



③ Create truth table for $Y = A \oplus B$ and verify.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Key Takeaway

Boolean Algebra simplifies and analyzes digital circuits. Mastering it helps in designing efficient logic circuits and solving real-world problems.

Applications

- ☑ Digital Circuit Design
- ☑ Embedded Systems
- ☑ Computer Architecture
- ☑ Microcontrollers
- ☑ Signal Processing
- ☑ Communication Systems

Tips

- ☑ Practice truth tables.
- ☑ Use identities to simplify.
- ☑ Draw circuits to verify.
- ☑ Understand each gate deeply.

Remember

- ☑ Logic gates are the building blocks of all digital systems.
- ☑ Practice + Visualization = Mastery!

17. Algebraic Structures

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What are Algebraic Structures?

Algebraic structures are sets equipped with one or more operations that satisfy certain axioms (laws). They provide a mathematical framework to model and solve problems.

Notation

- $(S, *)$: Set S with operation $*$
- $a, b, c \in S$: Elements of S
- $*$: Binary operation $S \times S \to S$
- e : Identity element (if exists)
- a^{-1} : Inverse of a (if exists)

Basic Properties (for operation $*$)

Property	Meaning
1 Closure	$a * b \in S$, for all $a, b \in S$
2 Associative	$(a * b) * c = a * (b * c)$
3 Commutative	$a * b = b * a$
4 Identity	$\exists\, e \in S$ such that $e * a = a * e = a$
5 Inverse	$\forall\, a \in S, \exists\, a^{-1} \in S$ such that $a * a^{-1} = a^{-1} * a = e$
6 Distributive	$a * (b + c) = a * b + a * c$ (and $(a + b) * c = a * c + b * c$)

Main Algebraic Structures

- Semigroup : $(S, *)$
  Associative
- Monoid : $(S, *, e)$
  Associative + Identity
- Group : $(S, *, e)$
  Associative + Identity + Inverse
- Abelian (Commutative) Group :
  Group + Commutative
- Ring : $(R, +, \cdot)$
  $(R, +)$ is Abelian group
  $(R, \cdot)$ is associative
  $\cdot$ is distributive over $+$
- Field : $(F, +, \cdot)$
  Ring + $(F \setminus \{0\}, \cdot)$ is Abelian group

— Structures at a Glance —

Structure	Notation	Axioms / Requirements	Example	Not Example
Semigroup	$(S, *)$	Closure, Associative	$(\mathbb{Z}, +)$, $(\mathbb{Z}, \cdot)$	$(\mathbb{Z}, -)$ not closed
Monoid	$(S, *, e)$	Semigroup + Identity	$(\mathbb{N}, +, 0)$, (String*, concatenation, ε)	$(\mathbb{N}, \cdot)$ no identity
Group	$(S, *, e)$	Monoid + Inverse	$(\mathbb{Z}, +, 0)$, $(\mathbb{R} - \{0\}, \cdot, 1)$	$(\mathbb{N}, +)$ no inverse
Abelian Group	$(S, *, e)$	Group + Commutative	$(\mathbb{Z}, +, 0)$, $(\mathbb{R}, +, 0)$	(Matrices under $\cdot$)
Ring	$(R, +, \cdot)$	$(R, +)$ Abelian Group $(R, \cdot)$ Semigroup	$(\mathbb{Z}, +, \cdot)$, $(M_n(\mathbb{R}), +, \cdot)$	$(\mathbb{Z}, \cdot)$ (no additive inverse)
Field	$(F, +, \cdot)$	Ring + $(F \setminus \{0\}, \cdot)$ Abelian Group	$(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$	$(\mathbb{Z}, +, \cdot)$ (no multiplicative inverse for non-zero elements)

1. Semigroup

A set S with a binary operation $*$ satisfying closure and associativity.

Axioms

1. Closure
2. $(a * b) * c = a * (b * c)$
   $\forall\, a, b, c \in S$

Example

$(\mathbb{Z}, +)$ is a semigroup.

2. Monoid

A semigroup with an identity element e.

Axioms

1. Closure
2. Associative
3. $\exists\, e \in S, e * a = a * e = a$
   $\forall\, a \in S$

Example

$(\mathbb{N}, +, 0)$ is a monoid.

3. Group

A monoid in which every element has an inverse.

Axioms

1. Closure
2. Associative
3. Identity (e)
4. Inverse (a^{-1})

Example

$(\mathbb{Z}, +, 0)$ is a group.

4. Abelian (Commutative) Group

A group in which the operation is commutative.

Axioms

1. Group axioms
2. $a * b = b * a \quad \forall\, a, b \in S$

Example

$(\mathbb{R}, +, 0)$ is an abelian group.

5. Ring

Two operations $+$ and $\cdot$ satisfying ring axioms.

Axioms

1. $(R, +)$ is abelian group
2. $(R, \cdot)$ is semigroup
3. $a \cdot (b + c) = a \cdot b + a \cdot c$
   $(a + b) \cdot c = a \cdot c + b \cdot c$

Example

$(\mathbb{Z}, +, \cdot)$ is a ring.

6. Field

A ring in which every non-zero element has a multiplicative inverse.

Axioms

1. Ring axioms
2. $(F \setminus \{0\}, \cdot)$ is abelian group

Example

$(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ are fields.

— Common Examples —

Structure	Set S	Operation(s)	Identity	Notes
$(\mathbb{Z}, +)$	Integers	Addition	0	Abelian Group
$(\mathbb{Z}_n, +_n)$	Integers mod n	Addition mod n	0	Abelian Group
$(\mathbb{Z}_n^*, \cdot_n)$	Integers mod n	Multiplication mod n	1	Abelian Group (if n prime)
$(\mathbb{N}, +)$	Natural numbers	Addition	0	Monoid (not group)
$(\mathbb{N}, \cdot)$	Natural numbers	Multiplication	1	Monoid (not group)
$(M_n(R), +, \cdot)$	$n \times n$ matrices	Matrix $+$, Matrix $\cdot$	Zero matrix, Identity matrix	Ring (not field)
$(\mathbb{R}, +, \cdot)$	Real numbers	Usual $+$, $\cdot$	0, 1	Field

Some Useful Facts

- Every group is a monoid and semigroup.
- Every abelian group is a group.
- Every field is a ring but not every ring is a field.
- $(\mathbb{Z}_n, +_n)$ is an abelian group for any n.
- $(\mathbb{Z}_n^*, \cdot_n)$ is an abelian group iff n is prime.
- Matrices under multiplication form a semigroup but usually not a group.

Substructures

- Subsemigroup: Non-empty subset closed under $*$.
- Submonoid: Subsemigroup containing identity.
- Subgroup: Submonoid in which every element has inverse in the subset.
- Subring: Subset closed under $+$, $\cdot$ and containing additive inverses.
- Subfield: Subring in which every non-zero element has multiplicative inverse in the subset.

Quick Property Summary —

Structure	Closure	Associative	Identity	Inverse	Commutative
Semigroup	✓	✓	✗	✗	Maybe
Monoid	✓	✓	✓	✗	Maybe
Group	✓	✓	✓	✓	Maybe
Abelian Group	✓	✓	✓	✓	✓
Ring	✓	✓	✓	✓	(+) Yes (·) No
Field	✓	✓	✓	✓	(+) Yes (·) Yes (except 0)

Example Problem

Q. Let $(\mathbb{Z}_6, +_6)$ be the set of integers modulo 6 under addition modulo 6.

1. Is it a group?
2. Find the inverse of 2.

Solution :

1. $(\mathbb{Z}_6, +_6)$ is an abelian group.
2. Inverse of 2 is 4 since $2 +_6 4 = 0$.
   (Because $2 + 4 = 6 \equiv 0 \pmod 6$)

Key Takeaway

Algebraic structures help us understand the properties of operations on sets. They are the foundation of abstract algebra and many areas of computer science and mathematics.

Applications

- ☑ Cryptography (Groups, Fields)
- ☑ Coding Theory (Finite Fields)
- ☑ Automata Theory (Semigroups)
- ☑ Computer Graphics (Matrices)
- ☑ Network Theory, Algorithms
- ☑ Databases & Formal Verification

Remember

- ☑ Always verify closure first.
- ☑ Check associativity carefully.
- ☑ Identify identity and inverse.
- ☑ Don't forget distributive law in rings.
- ☑ Practice with small examples!

Notation Cheat Sheet

- $\mathbb{Z}$: Integers
- $\mathbb{N}$: Natural numbers
- $\mathbb{Q}$: Rational numbers
- $\mathbb{R}$: Real numbers
- $\mathbb{C}$: Complex numbers
- $\mathbb{Z}_n$: Integers modulo n
- $a +_n b$: $(a + b) \bmod n$
- $a \cdot_n b$: $(a \cdot b) \bmod n$

18. Number Theory & Modular Arithmetic

What is Number Theory?

Number Theory deals with the properties of integers. It is the foundation for many areas in computer science like algorithms, cryptography, hashing, etc.

Notation

- $\mathbb{N}$: Set of natural numbers $\{1, 2, 3, \ldots\}$
- $\mathbb{Z}$: Set of integers $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$
- $a \mid b$: a divides b
- $\gcd(a, b)$: Greatest Common Divisor
- $\text{lcm}(a, b)$: Least Common Multiple
- $a \equiv b \pmod m$: a is congruent to b modulo m

Basic Number Theory

- Divisibility Rules

Divisible by	Rule
2	Last digit is divisible by 2.
3	Sum of digits is divisible by 3.
4	Last two digits form a number divisible by 4.
5	Last digit is 0 or 5.
6	Divisible by both 2 and 3.
9	Sum of digits is divisible by 9.
10	Last digit is 0.

- **Prime Number:** A number > 1 that has exactly two distinct positive divisors: 1 and itself. (2, 3, 5, 7, 11, ...)
- **Composite Number:** A number > 1 that is not prime.
- **Fundamental Theorem of Arithmetic:** Every integer > 1 can be written uniquely as a product of prime numbers (up to order).

Important Formulas

1. $\gcd(a, b) \cdot \text{lcm}(a, b) = |a \cdot b|$
2. $\gcd(a, b) = \gcd(b, a \bmod b)$ (Euclid's Algorithm)
3. If p is prime and $p \mid ab$ then $p \mid a$ or $p \mid b$.
4. If $\gcd(a, b) = 1$ and $a \mid bc$ then $a \mid c$.

Euclid's Algorithm (GCD)

To find $\gcd(a, b)$:

1. If $b = 0$, return a.
2. Else return $\gcd(b, a \bmod b)$.

Example: Find $\gcd(48, 18)$

$\gcd(48, 18) = \gcd(18, 48 \bmod 18)$

$= \gcd(18, 12) = \gcd(12, 6)$

$= \gcd(6, 0) = 6$

Least Common Multiple (LCM)

- $\text{lcm}(a, b) = \dfrac{|a \cdot b|}{\gcd(a, b)}$

Generalize for more than two numbers.

Example: $\text{lcm}(12, 18)$

$\gcd(12, 18) = 6$

$\Rightarrow \text{lcm}(12, 18) = \dfrac{12 \times 18}{6} = 36$

Prime Related Theorems

- Number of primes is infinite.
- Every integer > 1 is either prime or can be expressed as a product of primes.
- For any $n > 1$, number of primes $\leq n$ is approximately $\dfrac{n}{\ln n}$ (Prime Number Theorem).

Modular Arithmetic (Basics)

For integers a, b, m $(m > 0)$:

$a \equiv b \pmod m$ if $m \mid (a - b)$

(Meaning a and b leave same remainder when divided by m.)

Example: $17 \equiv 5 \pmod{12}$ (since $17 - 5 = 12$)

Rules of Modular Arithmetic

Operation	Rule (for any integers a, b, c and $m > 0$)
1. Addition	$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$
2. Subtraction	$(a - b) \bmod m = ((a \bmod m) - (b \bmod m) + m) \bmod m$
3. Multiplication	$(a \cdot b) \bmod m = ((a \bmod m) \cdot (b \bmod m)) \bmod m$
4. Division	$a/b \bmod m = (a \cdot b^{-1}) \bmod m$ (if b^{-1} exists)
5. Power	$(a^k) \bmod m = ((a \bmod m)^k) \bmod m$
6. Distributive	$(a(b + c)) \bmod m = ((ab) + (ac)) \bmod m$
7. Negation	$(-a) \bmod m = (m - (a \bmod m)) \bmod m$
8. Congruence	If $a \equiv b \pmod m$ and $c \equiv d \pmod m$ then $a + c \equiv b + d \pmod m$, $ac \equiv bd \pmod m$

Modular Exponentiation (Fast Power)

To compute $a^n \bmod m$ efficiently.

Use Binary Exponentiation.

Example:

Compute $3^{13} \bmod 17$

$13_{(10)} = 1101_2$

$3^{13} = 3^8 \cdot 3^4 \cdot 3^1$

$3^2 \equiv 9$, $3^4 \equiv 81 \equiv 13$,

$3^8 \equiv 13^2 \equiv 169 \equiv 16 \pmod{17}$

$3^{13} \equiv 16 \cdot 13 \cdot 3 \equiv 624 \equiv 12$

So, $3^{13} \bmod 17 = 12$

Modular Multiplicative Inverse

$b^{-1} \bmod m$ is the number x such that $b \cdot x \equiv 1 \pmod m$.

Exists iff $\gcd(b, m) = 1$.

Find using Extended Euclid Algorithm.

Example:

Find inverse of 3 modulo 11.

Since $3 \cdot 4 = 12 \equiv 1 \pmod{11}$,

$3^{-1} \equiv 4 \pmod{11}$.

Fermat's Little Theorem

If p is prime and a is not divisible by p, then:

$a^{p-1} \equiv 1 \pmod p$

Or equivalently, $a^p \equiv a \pmod p$.

Example:

For $p = 7$, $a = 3$:

$3^6 \equiv 1 \pmod 7$ ✓

Euler's Totient Function $\varphi(n)$

$\varphi(n)$ = number of integers in $\{1, 2, \ldots, n\}$ that are coprime to n.

Some values:

$\varphi(p) = p - 1$ (p is prime)

$\varphi(p^k) = p^k - p^{k-1}$

$\varphi(ab) = \varphi(a)\varphi(b)$ if $\gcd(a, b) = 1$

Euler's Theorem

If $\gcd(a, n) = 1$, then:

$a^{\varphi(n)} \equiv 1 \pmod n$

Corollary (Generalized Fermat):

If n is prime, Euler's Theorem reduces to Fermat's Little Theorem.

Common Examples

1. Check Divisibility

Is 123456 divisible by 9?

Sum of digits = 1+2+3+4+5+6 = 21

21 is divisible by 9?

No. So, 123456 is not divisible by 9.

2. GCD & LCM

Find GCD and LCM of 84 and 126.

$\gcd(84, 126) = 42$

$\text{lcm}(84, 126) = \dfrac{84 \times 126}{42} = 252$

3. Solve Congruence

Solve $2x \equiv 10 \pmod{17}$.

Inverse of 2 mod 17 is 9

$(2 \cdot 9 = 18 \equiv 1)$

$x \equiv 10 \cdot 9 \equiv 90 \equiv 5 \pmod{17}$

So, $x = 5$.

4. Modular Exponentiation

Compute $5^{117} \bmod 19$.

$117 \equiv 5 \pmod{18}$ (by Fermat, 19 prime)

$5^{18} \equiv 1 \pmod{19}$

$\Rightarrow 5^{117} \equiv 5^5 \pmod{19}$

$5^2 = 25 \equiv 6$, $5^4 \equiv 6^2 = 36 \equiv 17$

$5^5 = 5^4 \cdot 5 \equiv 17 \cdot 5 = 85 \equiv 9$

So, $5^{117} \bmod 19 = 9$

5. Inverse using Extended Euclid

Find inverse of 7 modulo 26.

Using extended Euclid:

$26 = 3 \cdot 7 + 5$

$7 = 1 \cdot 5 + 2$

$5 = 2 \cdot 2 + 1$

$1 = 5 - 2 \cdot 2 = 5 - 2(7 - 5) = 3 \cdot 5 - 2 \cdot 7$

$= 3(26 - 3 \cdot 7) - 2 \cdot 7 = 3 \cdot 26 - 11 \cdot 7$

$-11 \cdot 7 \equiv 1 \pmod{26}$

Inverse of 7 mod 26 is $-11 \equiv 15$.

Sieve of Eratosthenes (Find all primes $\leq n$)

1. Create array of size $n+1$ and mark all as prime.
2. Mark 0 and 1 as non-prime.
3. For $i = 2$ to $\sqrt{n}$:
   If prime[i] is true, mark all multiples of i as non-prime.
4. Remaining numbers are prime.

Example: Primes ≤ 30

2, 3, 5, 7, 11, 13, 17, 19, 23, 29

Chinese Remainder Theorem (CRT)

If $x \equiv a_1 \pmod{m_1}$, $x \equiv a_2 \pmod{m_2}$, ..., $x \equiv a_k \pmod{m_k}$

and $m_1, m_2, \ldots, m_k$ are pairwise coprime,

then there exists a unique solution modulo $M = m_1 m_2 \ldots m_k$.

Example:

$x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 7$

Solution: $x = 23 \pmod{105}$ (Check: $23 \% 3 = 2$, $23 \% 5 = 3$, $23 \% 7 = 2$)

Some Useful Identities

- $n \mid (a^n - a)$
- $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \ldots + b^{n-1})$
- $\displaystyle\sum_{i=0}^{n-1} i = \frac{n(n-1)}{2}$
- $1 + 2 + \ldots + n = \dfrac{n(n+1)}{2}$
- $1 + 2^0 + 2^1 + \ldots + 2^{n-1} = 2^n - 1$

Key Takeaway

Number Theory and Modular Arithmetic are the backbone of many powerful algorithms and real-world applications like cryptography, hashing, and security!

Applications

- ☑ Cryptography (RSA, Diffie-Hellman)
- ☑ Hashing & Checksums
- ☑ Random Number Generation
- ☑ Competitive Programming
- ☑ Error Detection & Correction Codes

Tips

- ☑ Practice GCD, LCM and Congruences
- ☑ Master Fast Exponentiation
- ☑ Remember little theorems (Fermat, Euler)
- ☑ Solve problems daily!

Remember

- ☑ Mathematics is the language of logic.
- ☑ The more you practice, the stronger your algorithms!

19. Important Numerical + Exam Problems

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Why Practice Numerical ?

- Improves speed & accuracy
- Helps in clearing coding & technical rounds
- Strengthens problem solving skills
- Frequently asked in interviews & competitive exams

Tips to Solve Quickly

- ☑ Read the question carefully.
- ☑ Identify what is asked.
- ☑ Use formulas / properties smartly.
- ☑ Do rough work neatly.
- ☑ Check edge cases & constraints.
- ☑ Verify the answer.

Common Formulas (Quick Reference)

- Sum of first n natural numbers = $n(n+1)/2$
- Sum of first n squares = $n(n+1)(2n+1)/6$
- Sum of first n cubes = $[n(n+1)/2]^2$
- $nCr = n! / (r!(n-r)!$
- Permutations: $nPr = n! / (n-r)!$
- Number of digits in $n = [\log_{10} n] + 1$
- $GCD(a,b) \times LCM(a,b) = a \times b$
- $a^m \bmod n \times a^n \bmod n = a^{m+n} \bmod n$



— IMPORTANT NUMERICAL PROBLEMS —

1. Basics

1. Find the sum of first 50 natural numbers.

   Ans. 1275

2. How many digits are there in 2^{1000} ?

   Ans. 302

2. Divisibility

1. Is 123456789 divisible by 9?

   Ans. Yes

2. Find the smallest 5-digit number divisible by 24.

   Ans. 10008

3. GCD & LCM

1. Find GCD and LCM of 84 and 126.

   Ans. GCD = 42, LCM = 252

2. If GCD(a,b) = 18 and LCM(a,b) = 180, and a = 36. Find b.

   Ans. 90

4. Primes

1. Find the 10th prime number.

   Ans. 29

2. How many primes are there between 1 and 100?

   Ans. 25

5. Modular Arithmetic

1. What is $2024^{2024} \bmod 100$?

   Ans. 76

2. Find the remainder when (987654321) is divided by 97.

   Ans. 65

6. Exponents

1. Simplify: $(2^5 \times 2^3) / 2^4$

   Ans. 16

2. Find the last digit of 7^{1234}.

   Ans. 1

7. Permutations & Combinations

1. Find $10P_3$.

   Ans. 720

2. How many ways can we choose 3 people from 12?

   Ans. 220

8. Probability

1. A coin is tossed 3 times. Find probability of getting exactly 2 heads.

   Ans. 3/8

2. A box has 4 red, 5 blue and 6 green balls. Find the probability of drawing a blue ball.

   Ans. 5/15 = 1/3

9. Sets

1. If $n(A) = 30$, $n(B) = 20$ and $n(A \cap B) = 10$. Find $n(A \cup B)$.

   Ans. 40

2. If $n(U) = 100$, $n(A) = 40$, $n(B) = 30$, $n(A \cup B) = 50$. Find $n((A \cup B)')$.

   Ans. 50

10. Sequences & Series

1. Find the 20th term of AP: 3, 7, 11, 15, ...

   Ans. 79

2. Find the sum of first 20 terms of GP: 2, 6, 18, ...

   Ans. $2(3^{200}-1)/(3-1) = 1747635762$

11. Number System

1. Convert $(1011011)_2$ to decimal.

   Ans. 91

2. Convert $(3AF)_{16}$ to decimal.

   Ans. 943

12. Bit Manipulation

1. Find the value of (5 & 3), (5 | 3), (5 ^ 3).

   Ans. 1, 7, 6

2. If n = 10 $(1010)_2$. Find n << 2.

   Ans. 40 $(101000)_2$

13. Counting Problems

1. How many 4-digit numbers can be formed using digits 1, 2, 3, 4 without repetition?

   Ans. 24

2. How many ways can we arrange the letters of "BANANA"?

   Ans. 60

14. Recurrence Relations

1. Solve: $T(n) = T(n-1) + 2$, $T(1) = 1$.

   Ans. $T(n) = 2n - 1$

2. Solve: $T(n) = 2T(n/2) + n$, $T(1) = 1$.

   Ans. $T(n) = n \log_2 n$

15. Graph Theory

1. In a complete graph with 10 vertices, how many edges are there?

   Ans. 45

2. In a tree with n vertices, how many edges are there?

   Ans. n - 1

MIXED ADVANCED PROBLEMS (EXAM LEVEL)

1. Find the smallest positive integer n such that n is divisible by 6, 8, 12 and 15.

   Ans. 120

2. If $a \equiv 3 \pmod 5$ and $b \equiv 2 \pmod 5$, find $(3a + 4b) \bmod 5$.

   Ans. 1

3. In how many ways can the letters of the word "COMPUTER" be arranged so that vowels never come together?

   Ans. 30240

4. A fair die is rolled 6 times. Find the probability that the sum of numbers is 21.

   Ans. 1/6

5. Solve for n: nC2 + nC3 = 45

   Ans. n = 10

6. How many solutions (x, y) in non-negative integers are there for the equation: x + 2y + 3z = 12?

   Ans. 28 (where $z \geq 0$)

★ Key Takeaway

Regular practice of numericals improves confidence and speed. During exams / interviews, smart approach + accuracy = selection!



Where These Are Asked

- ☑ Campus Placements
- ☑ Product Company Interviews
- ☑ Competitive Programming
- ☑ GATE / CAT / GRE / Other Exams
- ☑ Coding Contests

Time-Saving Tricks

- Memorize important formulas
- Break problem into small parts
- Check for patterns & properties
- Use estimation to avoid lengthy calculations

Remember

- ☑ Practice daily
- ☑ Analyse mistakes
- ☑ Track time
- ☑ Consistency is the key

Practice More → Think Smart → Solve Fast → Succeed Always!



20. TOP 100 DISCRETE MATHEMATICS QUESTIONS + QUICK REVISION CHEAT SHEET

TOP 100 IMPORTANT QUESTIONS

1 Logic & Propositional Logic (1–10)

1. What is a proposition?
2. What are truth values?
3. Negation of a proposition.
4. Conjunction of two propositions.
5. Disjunction of two propositions.
6. Implication of two propositions.
7. Biconditional of two propositions.
8. Converse, inverse, contrapositive.
9. Tautology and contradiction.
10. Convert to CNF or DNF.

2 Predicate Logic (11–20)

11. What is a predicate?
12. Variables vs. constants.
13. Universal quantifier (∀).
14. Existential quantifier (∃).
15. Negation of quantified statements.
16. Free vs bound variables.
17. Predicate logic to English.
18. English to predicate logic.
19. Validity of arguments.
20. Rules of inference.

3 Sets (21–30)

21. What is a set?
22. Types of sets.
23. Subset and proper subset.
24. Power set of a set.
25. Union of two sets.
26. Intersection of two sets.
27. Complement of a set.
28. Set difference.
29. De Morgan's laws for sets.
30. Venn diagram problems.

4 Relations (31–40)

31. What is a relation?
32. Domain and codomain.
33. Range of a relation.
34. Types of relations.
35. Reflexive relation.
36. Symmetric relation.
37. Antisymmetric relation.
38. Transitive relation.
39. Equivalence relation.
40. Composition of relations.

5 Functions (41–50)

41. What is a function?
42. Domain, codomain, range.
43. One-to-one function.
44. Onto function.
45. Bijective function.
46. Inverse of a function.
47. Composition of functions.
48. Types of functions.
49. Count of functions from A to B.
50. Properties of functions.

6 Mathematical Induction (51–60)

51. Principle of mathematical induction.
52. Prove $1 + 2 + \ldots + n = n(n+1)/2$.
53. Prove $1^2 + 2^2 + \ldots + n^2 = n(n+1)(2n+1)/6$.
54. Prove $2^n > n$ for $n \geq 5$.
55. Prove divisibility using induction.
56. Strong induction.
57. Structural induction – basics.
58. Induction on inequalities.
59. Incorrect induction – find flaw.
60. Real world induction problems.

7 Counting (61–70)

61. Addition principle.
62. Multiplication principle.
63. Factorial – definition.
64. Permutations (nPr).
65. Combinations (nCr).
66. nCr properties.
67. Circular permutations.
68. Multiset permutations.
69. Pigeonhole principle.
70. Inclusion–Exclusion principle.

8 Graph Theory Basics (71–80)

71. What is a graph?
72. Types of graphs.
73. Degree of a vertex.
74. Handshaking lemma.
75. Walk, trail, path, cycle.
76. Connected graph.
77. Euler path and circuit.
78. Hamilton path and cycle.
79. Bipartite graph.
80. Planar graph.

9 Trees (81–90)

81. What is a tree?
82. Properties of trees
83. Leaves in a tree.
84. Number of edges in a tree.
85. Degree sum property.
86. Binary tree basics.
87. Full, complete, perfect tree.
88. Spanning tree.
89. Minimum spanning tree.
90. Applications of trees.

10 Discrete Structures & Others (91–100)

91. What is a semigroup?
92. Monoid – definition and example.
93. Group – definition and example.
94. Ring – definition and example.
95. Field – definition and example.
96. Lattice – definition.
97. Poset and Hasse diagram.
98. Recurrence relation – solve.
99. Generating functions – basics.
100. Applications of discrete math.

PRACTICE SMART – SCORE HIGH!

TIPS TO ATTEMPT QUESTIONS

- ☑ Read the question carefully and identify the concept.
- ☑ Write given information and what is asked.
- ☑ Use definitions, theorems and properties.
- ☑ Draw diagrams / tables wherever helpful.
- ☑ Verify your answer with a small example.
- ☑ Manage time – easy first, tough later.

Logic
Work with truth values and logical connectives.

Sets
Union (∪), Intersection (∩), Complement (')

Relations
Reflexive, Symmetric, Antisymmetric, Transitive.

Functions
One-One, Onto, Bijective, Composition.

QUICK REVISION CHEAT SHEET

1. LOGIC LAWS

Idempotent	$p \wedge p = p$, $p \vee p = p$
Domination	$p \wedge T = p$, $p \vee F = p$
Identity	$p \wedge T = p$, $p \vee F = p$
Commutative	$p \wedge q = q \wedge p$ $p \vee q = q \vee p$
Associative	$(p \wedge q) \wedge r = p \wedge (q \wedge r)$ $(p \vee q) \vee r = p \vee (q \vee r)$
Distributive	$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
De Morgan	$(p \wedge q)' = p' \vee q'$ $(p \vee q)' = p' \wedge q'$
Double Negation	$(p')' = p$
Absorption	$p \vee (p \wedge q) = p$ $p \wedge (p \vee q) = p$
Implication	$p \to q \equiv p' \vee q$
Biconditional	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

2. TRUTH TABLES

p	q	¬p	p∧q	p∨q	p→q	p↔q
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

3. SET LAWS

- $A \cup A = A$, $A \cap A = A$
- $A \cup \emptyset = A$, $A \cap U = A$
- $A \cup A^c = U$, $A \cap A^c = \emptyset$
- $(A \cup B)^c = A^c \cap B^c$
- $(A \cap B)^c = A^c \cup B^c$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

4. QUANTIFIERS

$\forall x\ P(x)$: For all x, P(x) is true.

$\exists x\ P(x)$: There exists at least one x such that P(x) is true.

$\neg\ \forall x\ P(x) \equiv \exists x\ \neg P(x)$

$\neg\ \exists x\ P(x) \equiv \forall x\ \neg P(x)$

5. RELATIONS

- Reflexive: $\forall a \in A,\ (a, a) \in R$
- Symmetric: $\forall a, b \in A,\ (a, b) \in R \Rightarrow (b, a) \in R$
- Antisymmetric: $\forall a, b \in A,\ (a, b) \in R \wedge (b, a) \in R \Rightarrow a = b$
- Transitive: $\forall a, b, c \in A,\ (a, b) \in R \wedge (b, c) \in R \Rightarrow (a, c) \in R$

6. FUNCTIONS

- One-One: $f(a) = f(b) \Rightarrow a = b$
- Onto: $\forall y \in$ Codomain, $\exists x \in$ Domain such that $f(x) = y$
- Bijective: One-One and Onto
- $f^{-1}(f(x)) = x$, $f(f^{-1}(y)) = y$

7. COUNTING FORMULAS

- $n! = n(n-1) \ldots 2 \cdot 1$, $0! = 1$
- $nP_r = \dfrac{n!}{(n-r)!}$
- $nC_r = \dfrac{n!}{r!(n-r)!}$
- $nC_r = nC_{n-r}$
- $\sum_{r=0}^{n} nC_r = 2^n$

8. IMPORTANT THEOREMS

- Principle of Mathematical Induction
  Base case, Inductive step
- Pigeonhole Principle
  If n+1 objects in n boxes, then some box has at least 2 objects.
- Handshaking Lemma
  $\sum_{v \in V} \deg(v) = 2|E|$
- Inclusion–Exclusion Principle
  $|A \cup B| = |A| + |B| - |A \cap B|$
  (extends to 3 or more sets)

9. GRAPH THEORY FORMULAS

- For undirected graph: $\sum \deg(v) = 2|E|$
- In a tree with n vertices: $|E| = n - 1$
- Number of edges in complete graph K_n: $|E| = \dfrac{n(n-1)}{2}$
- Minimum edges for connected graph: $n - 1$
- Maximum edges in simple graph: $\dfrac{n(n-1)}{2}$
- For planar graph: $V - E + F = 2$

10. ALGEBRAIC STRUCTURES – QUICK VIEW

Structure	Set	Operation	Key Properties
Semigroup	S	*	Closure, Associative
Monoid	S	*	Closure, Associative, Identity e
Group	G	*	Monoid + Inverse for every element
Abelian Group	G	+	Group + Commutative
Ring	R	+, ·	Closure, Assoc. (both), Distributive, Identities 0, 1
Field	F	+, ·	Ring + Every non-zero element has a multiplicative inverse

COMMON SYMBOLS

∀	For all
∃	There exists
∈	Element of
⊆	Subset
⇒	Implies
⇔	Equivalent
∅	Empty set

N, Z, Q, R
Natural, Integer, Rational, Real numbers

ONE LINE REVISION

Counting
nCr
Permutations, Combinations, Pigeonhole, Inclusion-Exclusion.

Graphs
Vertices (V), Edges (E), Degree, Paths, Trees, MST.

Graphs
Σ
Groups, Rings, Fields, Fields.

Algebra
Groups, (VI, Crelás, Rings, Posets.

Induction
Base case + Inductive step = Proof!

☆ Practice Consistently • Revise Regularly • Solve Smartly = Master Discrete Mathematics!